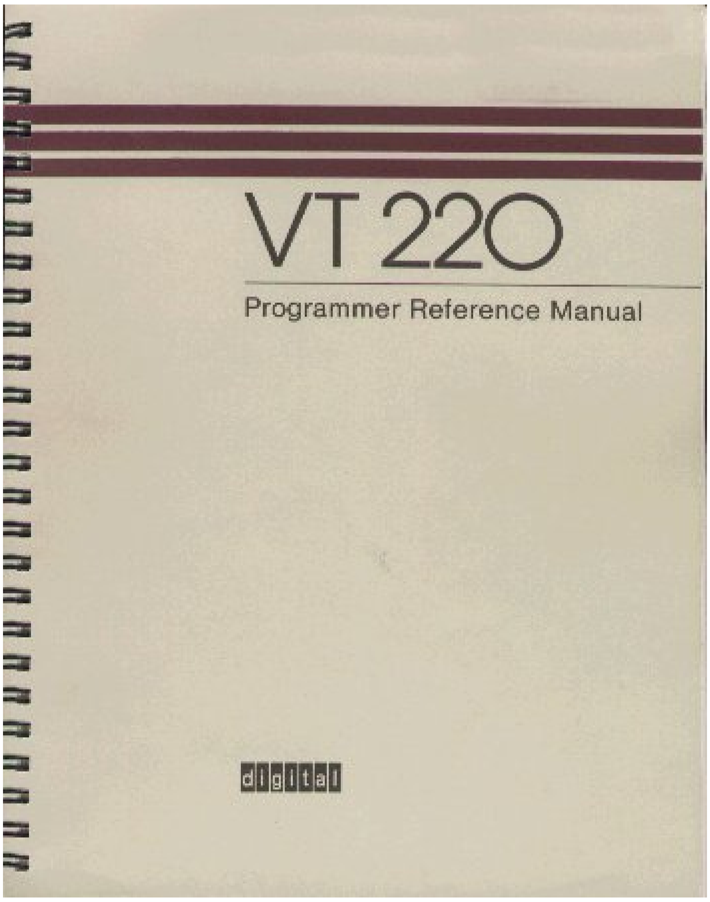
VT220 Programmer Reference Manual



This document is a reproduction of the second edition of this book, published by Digital in August 1984. The first edition was published in August 1983. Although I have proof-read this, there may still be errors remaining. If you find one, please tell me!

Thanks to Hellmuth Michaelis for giving me a copy of this book.

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Introduction

This reference manual is for people with a knowledge of basic computer programming. The manual provides the information needed to access VT220 features. The manual is organized into the following chapters and appendices.

Chapter 1, "Terminal Overview", introduces you to the terminal. The chapter briefly discusses the terminal's capabilities and operating modes.

Chapter 2, "Character Encoding", describes the character encoding schemes used for the terminal. The chapter also describes the terminal's character sets and provides an overview of the control function format.

Chapter 3, "Transmitted Codes", describes the codes that the terminal sends to a program.

Chapter 4, "Received Codes", describes the codes that the terminal recognizes and responds to.

Appendix A, "VT220/VT102 Differences", describes the major differences between a VT102 terminal and the VT220 terminal operating in a VT100 mode.

Appendix B, "Additional VT220 Documents", describes other VT220 documents you can order from Digital.

Appendix C, "Using Control Codes", provides general suggestions on how to use VT220 control codes in any software application.

1 Terminal Overview

1.1 General

This chapter provides an overview of the VT220 terminal. The chapter describes the VT220's major features and operating modes.

The VT220 (Figure 1-1) is a general-purpose, video display terminal that uses ANSI standard functions. The terminal has two major components, a monitor/system unit and keyboard.

1.2 Terminal Features

The following sections summarize the major VT220 features.

1.2.1 Display Features

The VT220 uses a monochrome monitor with the following features.

- 24 rows × 80/132 columns
  7 × 10 dot matrix in a 10 × 10 cell for 80 columns
  7 × 10 dot matrix in a 9 × 10 cell for 132 columns

- 800 (horizontal) × 240 (vertical) pixels (80 columns)
  1188 (horizontal) × 240 (vertical) pixels (132 columns)

1.2.2 Text Features

This list summarizes the major VT220 text features.

- Five character sets of 94 characters each (including the DEC multinational character set)
- Down-line-loadable character set
- User-definable function keys
- Reverse video characters
- Underline characters
- Bold characters
- Blinking characters
- Double-height/double-width lines
- ANSI-compatible control functions
- VT52 mode

1.3 Communication Environment

The VT220 provides the following major communication features.

- Asynchronous communications to 19.2 Kbits per second
- EIA RS232C host port
- 20 mA host port
- EIA RS232C printer port
- 7-bit and 8-bit character formats

1.4 Operating States

The VT220 has three operating states.

- Set-up
- On-line
- Local

Set-Up
The set-up state lets you select or examine terminal operating features (such as transmit and receive speeds) from the keyboard. You can also use set-up to select the on-line and local states. The *VT220 Owner's Manual* describes set-up in detail.

You select set-up by pressing the Set-Up key.

On-Line
The on-line state lets the terminal communicate with a host computer. Data entered at the keyboard is sent to the host computer. Data received from the host computer is displayed on the monitor. You can also display data entered from the keyboard, if you select the local echo feature in set-up or with a control sequence.

You can only select on-line from set-up.

Local
The local state lets you place the host computer on "hold". Data entered at the keyboard is sent to the monitor, but not the host computer. Data received from the host computer is stored; this data is sent to the monitor after you put the terminal back on-line.

You can only select local from set-up.

1.5 Operating Modes

The VT220 has four major operating modes. You can select these modes from the keyboard, or from the host via control codes.

- VT200 mode, 7-bit controls
- VT200 mode, 8-bit controls
- VT100 mode
- VT52 mode

VT200 mode, 7-bit controls is the default mode and executes standard ANSI functions. This mode provides the full range of VT220 capabilities in an 8-bit communications environment with 7-bit controls (Chapter 2). This mode supports the DEC multinational character set or national replacement character (NRC) sets, depending on the character set mode selected. You can access both types of character sets from the keyboard, or from the host computer via control codes. This operating mode also provides some backward compatibility for existing VT100 software.

VT200 mode, 8-bit controls executes standard ANSI functions. This mode provides the full range of VT220 capabilities in an 8-bit communications environment with 8-bit controls (Chapter 2). This mode supports the DEC multinational character set or national replacement character (NRC) sets, depending on the character set mode selected. As in VT200 mode, 7-bit controls, you can access both types of character sets from the keyboard or with programmed control codes.

VT100 mode executes standard ANSI functions. It has a high degree of compatibility with Digital's VT102 terminal. (Appendix A describes VT220/VT102 differences). This mode restricts use of the keyboard to VT100 keys. All data is restricted to 7 bits, and only ASCII, national replacement characters (NRC), or special graphics characters are generated. This mode is provided for strict backward compatibility with existing software written for the VT100 terminal family.

VT52 mode executes Digital (DEC) private functions (not ANSI). It has a degree of compatibility with Digital's VT102 terminal operating in a VT52 mode. This mode restricts use of the keyboard to VT52 keys. All data is restricted to 7 bits, and only ASCII, U.K., or special graphics characters are generated.

1.6 Character Set Modes

The VT220 has two character set modes, multinational and national. You can select either mode from the keyboard in set-up, or from the host computer via control codes.

Multinational mode supports the DEC multinational character set (DEC MCS). The DEC MCS is an 8-bit character set that contains most characters used in the major European languages. The ASCII character set is included in the DEC MCS.

National mode supports the national replacement character sets (NRC sets). The NRC sets are a group of eleven 7-bit character sets. The national character set available is determined by the keyboard selected in set-up. Only one national character set is available for use at any one time. National mode restricts compatibility to a 7-bit environment in which the use of the DEC MCS is disabled.

2 Character Encoding

2.1 General

This chapter describes the character encoding concepts for the VT220 when operating in text mode. The chapter also describes the VT220 character sets and provides an overview of the control functions. You must have a basic understanding of the coding concepts described in this chapter before using the control functions described in Chapters 3 and 4.

2.2 Coding Standards

The VT220 uses an 8-bit character encoding scheme and a 7-bit code extension technique that are compatible with the following ANSI and ISO standards. ANSI (American National Standards Institute) and ISO (International Organization for Standardization) specify the current standards for character encoding used in the communications industry.

Standard	Description
ANSI X3.4 – 1977	American Standard Code for Information Interchange (ASCII)
ISO 646 – 1977	7-Bit Coded Character Set for Information Processing Interchange
ANSI X3.41 – 1974	Code Extension Techniques for Use with the 7-Bit Coded Character Set of American National Code Information Interchange
ISO Draft International Standard 2022.2	7-Bit and 8-Bit Coded Character Sets – Code Extension Techniques
ANSI X3.32 – 1973	Graphic Representation of the Control Characters of American National Code for Information Interchange
ANSI X3.64 – 1979	Additional Controls for Use with American National Standard for Information Interchange
ISO Draft International Standard 6429.2	Additional Control Functions for Character Imaging Devices

2.3 Code Table

A code table is a convenient way to represent 7-bit and 8-bit characters, because you can see groupings of characters and their relative codes clearly.

2.3.1 7-Bit ASCII Code Table

Table 2-1 is the 7-bit ASCII code table. There are 128 positions corresponding to 128 character codes, arranged in a matrix of 8 columns and 16 rows.

Each row represents a possible value of the four least significant bits of a 7-bit code (Figure 2-1). Each column represents a possible value of the three most significant bits.

Table 2-1 shows the octal, decimal, and hexadecimal code for each ASCII character. You can also represent any character by its position in the table. For example, the character H (column 4, row 8) can be represented as 4/8. This column/row notation is used to represent characters and codes throughout this manual. For example:

```
1/11 2/3 3/6
ESC   #   6
```

means that the ESC character is at column 1, row 11; the character # is at column 2, row 3; and the character 6 is at column 3, row 6.

The VT220 processes received characters based on two character types defined by ANSI, graphic characters and control characters.

Graphic characters are characters you can display on a video screen. The ASCII graphic characters are in positions 2/1 through 7/14 of Table 2-1. They include all American and English alphanumeric characters, plus punctuation marks and various text symbols. Examples are: C, n, ", !, +, and $ (the English pound sign is not an ASCII graphic character).

Control characters are not displayed. They are single-byte codes that perform specific functions in data communications and text processing. The ASCII control characters are in positions 0/0 through 1/15 (columns 0 and 1) of Table 2-1. The SP character (space, 2/0) may act as a graphic character or a control character, depending on the context. DEL (7/15) is always a control character.

Control character codes and functions are standardized by ANSI. Examples of ASCII control characters with their ANSI-standard mnemonics are: CR (carriage return), FF (form feed), CAN (cancel).

2.3.2 8-Bit Code Table

In general, the conventions for 7-bit character encoding also apply to 8-bit character encoding for the VT220. Table 2-2 shows the 8-bit code table. It has twice as many columns as the 7-bit table, because it contains 256 (versus 128) code values.

As with the 7-bit table, each row represents a possible value of the four least significant bits of an 8-bit code (Figure 2-2). Each column represents a possible value of the four most significant bits.

All codes on the left half of the 8-bit table (columns 0 through 7) are 7-bit compatible; their eighth bit is not set, and can be ignored or assumed to be 0. You can use these codes in a 7-bit or an 8-bit environment. All codes on the right half of the table (columns 8 through 15) have their eighth bit set. You can use these codes only in an 8-bit compatible environment.

The 8-bit code table (Table 2-2) has two sets of control characters, C0 (control zero) and C1 (control one). The table also has two sets of graphic characters, GL (graphic left) and GR (graphic right).

On the VT220, the basic functions of the C0 and C1 codes are defined by ANSI. C0 codes represent the ASCII control characters described earlier. The C0 codes are 7-bit compatible. The C1 codes represent 8-bit control characters that let you perform additional functions beyond those possible with the C0 codes. You can only use C1 codes directly in an 8-bit environment. Some C1 code positions are blank, because their functions are not yet standardized.

NOTE: The VT220 does not recognize all C0 and C1 codes. Chapter 4 identifies and describes the codes it does recognize; all others are simply ignored. (That is, no action is taken).

The GL and GR sets of codes are reserved for graphic characters. There are 94 GL codes in positions 2/1 through 7/14, and 94 GR codes in positions 10/1 through 15/14. By ANSI standards, positions 10/0 and 15/15 are not used. You can use GL codes in 7-bit or 8-bit environments. You can use GR codes only in an 8-bit environment.

2.4 Character Sets

You cannot change the functions of the C0 or C1 codes. However, you can map different sets of graphic characters into the GL and/or GR codes. The sets are stored in the terminal as a graphic repertoire. But you cannot use these graphics character sets until you map them into the GL or GR codes. Chapter 4 describes the commands for mapping graphic character sets into GL or GR.

The terminal's graphic repertoire consists of the following character sets, described in the following sections.

- DEC multinational (consists of the ASCII graphics set and DEC supplemental graphics set)
- DEC special graphics
- National replacement character (NRC) sets
- Down-line-loadable

2.4.1 DEC Multinational Character Set

By factory default, when you power up or reset the terminal, the DEC multinational character set (Table 2-3) is mapped into the 8-bit code matrix (columns 0 through 15).

The 7-bit compatible left half of the DEC multinational set is the ASCII graphics set. The C0 codes are the ASCII control characters, and the GL codes are the ASCII graphics set.

The 8-bit compatible right half of the DEC multinational set includes the C1 8-bit control characters in columns 8 and 9. The GR codes are the DEC supplemental graphics set. The DEC supplemental graphics set has alphabetic characters with accents and diacritical marks that appear in the major Western European alphabets. It also has other symbols not included in the ASCII graphics set.

The terminal can work with over a dozen national (Western European) keyboards. All keyboards assume the default DEC multinational character set mapping. The code descriptions in the rest of this manual also assume this mapping. Various characters from the DEC supplemental graphics set appear as standard (printing character) keys on different keyboards.

The DEC supplemental graphics character set is not available in VT52 and VT100 modes.

2.4.2 DEC Special Graphics Character Set

The terminal's graphic repertoire includes the DEC special graphics set (also known as the VT100 line drawing character set). This character set (Table 2-4) has about two-thirds of the ASCII graphic characters. It also has special symbols and short line segments. The line segments let you create a limited range of pictures while still using text mode.

Commands described in Chapter 4 let you map the DEC special graphics set into either GL or GR, replacing either the ASCII graphics set or the DEC supplemental graphics set. Digital recommends that you switch between ASCII and DEC special graphics in GL, because the latter has most of the ASCII graphic characters. Also, this mapping is compatible with a VT100 terminal.

2.4.3 National Replacement Character Sets (NRC sets)

The terminal's graphic character repertoire includes national replacement character sets (Tables 2-5 through 2-15). These sets are available when you select national mode. Only one national character set is available for use at any one time. THe NRC set used depends on the keyboard selection in set-up as outlined below.

Keyboard Selection	NRC Set
British	British
Danish	Norwegian/Danish
Dutch	Dutch
Finnish	Finnish
Flemish	French

Keyboard Selection	NRC Set
French/Belgian	French
French Canadian	French Canadian
German	German
Italian	Italian
Norwegian	Norwegian/Danish
Spanish	Spanish
Swedish	Swedish
Swiss (French)	Swiss
Swiss (German)	Swiss

2.4.4 Down-Line-Loadable Character Set

The terminal provides for a 94-character down-line-loadable graphic character set. You can define this character set and map it into either GL or GR, as described in Chapter 4. This feature is available only in VT200 mode.

2.5 Control Functions

You use control functions in your program to specify how the terminal should handle data. There are many uses for control functions. Here are some examples.

- Move the cursor on the display.
- Delete a line of text from the display.
- Change character and line attributes.
- Change graphic character sets.
- Set the terminal operating mode.

You can use all control functions in text mode and express them as single-byte or multibyte codes.

The single-byte codes are the C0 and C1 control characters. Your program can perform a limited number of functions using C0 characters. C1 characters give you a few more functions, but your program can use them directly only in an 8-bit environment.

Multibyte control codes represent far more functions, because they have more possible code combinations. These codes are called **escape sequences**, **control sequences**, and **device control strings**. Some sequences are ANSI standardized and used throughout the industry; others are private sequences created by manufacturers like Digital for specific families of products. Private sequences, like ANSI standardized sequences, follow ANSI standards for character code composition.

2.5.1 Escape Sequences

An escape sequence starts with the C0 character ESC (1/11), followed by one or more ASCII graphic characters. For example,

```
1/11 2/3 3/6
ESC   #   6
```

is an escape sequence that changes the current line of text to double-width characters.

Because escape sequences use only 7-bit characters, you can use them in 7-bit or 8-bit environments.

NOTE: When using escape or control sequences, remember that it is the code that defines a sequence -- not the graphic representation of the characters. The characters are shown for readability only and presume the DEC multinational character set mapping (ASCII graphics set in GL and DEC supplemental graphics set in GR).

One important use of escape sequences is extending the capability of 7-bit control functions. ANSI lets you use 2-byte escape sequences as 7-bit code extensions to express each of the C1 control codes. This is a valuable feature when your application must be compatible with a 7-bit environment. For example, the C1 characters CSI, SS3, and IND can be expressed as follows.

```
                    7-Bit Code Extension Equivalent
C1 Character         (Escape Sequence)

9/11                1/11  5/11
CSI                 ESC    [

8/15                1/11  4/15
SS3                 ESC    O

8/4                 1/11  4/4
IND                 ESC    D
```

In general, you can use the above code extension technique in two ways.

1. You can express any C1 control character as a 2-character escape sequence whose second character has a code that is 40 (hexadecimal) and 64 (decimal) less than that of the C1 character.
2. You can make any escape sequence whose second character is in the range of 4/0 through 5/15 one byte shorter by removing the ESC and adding 40 (hexadecimal) to the code of the second character. This generates an 8-bit control character.

2.5.2 Control Sequences

A control sequence starts with CSI (9/11), followed by one or more ASCII graphic characters. But CSI (9/11) can also be expressed as the 7-bit code extension ESC [(1/11, 5/11). So you can express all control sequences as escape sequences whose second character code is [(5/11). For example, the following two sequences are equivalent sequences that perform the same function. (They cause the display to use 132 columns per line rather than 80).

```
9/11 3/15 3/3 6/8
CSI   ?    3    h

1/11 5/11 3/15 3/3 6/8
ESC   [     ?    3   h
```

Whenever possible, you should use CSI instead of ESC [to introduce a control sequence. CSI uses one less byte than ESC [, so you gain processing speed. However, you can only use a sequence starting with CSI in an 8-bit environment (because CSI is a C1 control character).

2.5.3 Device Control Strings

A device control string (DCS) is a delimited string of characters used in a data stream as a logical entity for control purposes. It consists of an opening delimiter (a device control string introducer), a command string (data), and a closing delimiter (a string terminator).

You use device control strings to down-line-load character sets and definitions for user-defined keys.

The VT220 uses the following device control string format.

```
9/0            ..........       9/12
DCS            Data             ST

Device                          String
Control         .UDK            Terminator
String          .Character Set  (closing delimiter)
(opening
delimiter)
```

DCS is an 8-bit control character. You can also express it as ESC P (1/11, 5/0) when coding for a 7-bit environment.

ST is an 8-bit control character. You can also express it as ESC \ (1/11, 5/12) when coding for a 7-bit environment.

2.6 Working with 7-Bit and 8-Bit Environments

There are two requirements for using the terminal's 8-bit character set. Your program and communication environment must be 8-bit compatible, and the terminal must operate in a VT200 mode. When operating in VT100 or VT52 mode, you are limited to working in a 7-bit environment. The following sections describe conventions that apply in VT200 mode.

2.6.1 Conventions for Codes Received by the Terminal

The terminal expects to receive character codes in a form consistent with 8-bit coding. Your application can use the C0 and C1 control codes, as well as the 7-bit C1 code extensions, if necessary. The terminal always interprets these codes correctly. Chapter 4 describes all code extensions you may need to use, and their equivalent C1 control codes.

When your program sends GL or GR codes, the terminal interprets the codes according to the graphic character mapping currently in use. The factory-default mapping (which is set when you power up or reset the terminal) is the DEC multinational character set. This mapping assumes the current terminal mode is one of the VT200 modes.

2.6.2 Conventions for Codes Sent by the Terminal

Codes sent by the terminal to a program can come directly from the keyboard or in response to commands issued from the host (application program or operating system). In a VT200 mode, the terminal always sends all GL and GR graphic codes to the application exactly as they are generated, regardless of whether the application handles 8-bit codes correctly or not. If, however, a 7-bit communications line is used, C1 controls are sent as escape sequences and the terminal does not allow the generation of 8-bit graphic codes.

Most function keys on the keyboard generate multibyte control codes. Many of these codes start with either CSI (9/11) or SS3 (8/15), which are C1 characters. If your application environment cannot handle 8-bit codes, you can make the terminal automatically convert all C1 codes to their equivalent 7-bit code extensions before sending them to the application. To convert C1 codes, you use the DECSCL commands described in Chapter 4.

By default, the terminal is set to automatically convert all C1 codes sent to the application to 7-bit code extensions. However, to ensure the correct mode of operation, always use the appropriate DECSCL commands described in Chapter 4.

NOTE: New programs should accept both 7-bit and 8-bit forms of the C1 controls.

2.7 Display Controls Mode

The terminal has a display controls mode that lets you display control codes as graphic characters for debugging purposes (rather than executing them). You can select this mode by changing the "Interpret/Display Controls" field in the Set-Up Display screen. You cannot use an escape sequence or invoke this mode from the host computer.

When the terminal is in a VT200 mode, selecting the set-up "Display Controls" field temporarily loads C0, GL, C1, and GR as shown in Table 2-16. All characters are displayed in the font shown for C0, GL, C1, and GR.

When the terminal is in a VT52 or VT100 mode, selecting the set-up "Display Controls" field temporarily loads C0 and GL as shown in Table 2-16. All characters are displayed in the font shown for C0 and GL. (C1 and GR are meaningless in VT52 or VT100 modes).

When you select the "Display Controls" field, the terminal displays all control functions and prevents most from executing. There are only two exceptions. LF, FF, and VT cause a new line (CR LF), and XOFF (DC3) and XON (DC1) maintain flow control if enabled. LF, FF, and VT are displayed before CRLF is executed, and DC1 and DC3 are displayed after execution.

3 Transmitted Codes

3.1 General

This chapter describes the codes that the terminal sends to a program. The chapter assumes that you are familiar with the character encoding concepts described in Chapter 2.

Key codes generated in VT52 mode are listed if they differ from those generated in the ANSI-compatible (VT200 and VT100) modes.

The terminal can use 15 different national keyboards. This chapter describes significant differences among the keyboards.

3.2 Keyboard Codes

The terminal keyboard (Figure 3-1) consists of a main keypad, an editing keypad, an auxiliary keypad, and the top-row function keys.

3.2.1 Main Keypad

The main keypad consists of standard keys (used to generate letters, numbers, and symbols) and function keys (used to generate special function codes).

3.2.1.1 Standard Keys

The standard keys generate alphanumeric characters, either singly or in combination with other keys.

On the North American keyboard, the standard keys show only ASCII characters and generate only ASCII codes. There are no DEC supplemental characters among the standard keys. This is a special case, since most keyboards have some standard keys that generate DEC supplemental characters as well as ASCII characters.

The standard-key patterns vary among keyboards. Some graphic characters (either special symbols or characters with diacritical marks) may or may not be available as standard keys on particular keyboards. However, you can create any DEC multinational graphic character that is not available on a standard key, by typing a valid compose sequence.

Regardless of which keyboard you use and how you create a graphic character, each character is represented by a unique code. The code is based on the character's position in the code table. You can use all GL characters in both 7-bit (VT52 mode, VT100 mode, or 7-bit host line) and 8-bit (VT200 mode, 8-bit host line) environments. You can only use GR characters in an 8-bit environment.

All keyboards (except the North American Keyboard) have one or more standard keys that send different graphic characters (and corresponding codes). The graphic character sent depends on whether you selected "Typewriter Keys" or "Data Processing Keys" in the Keyboard Set-Up screen. (See Figure 3-2 for an example).

All keys affected by "Typewriter Keys" or "Data Processing Keys" have more than two characters shown on their keycaps. The key sends the character on the right side of the keycap when you select "Data Processing Keys", and the character on the left side of the keycap when you select "Typewriter Keys". You can select either shifted (upper) or unshifted (lower) character codes for these keys in the same way as for other nonalphabetic keys.

3.2.1.2 Function Keys

This section describes the main keypad function keys. The column/row notation used in the descriptions refers to Table 2-1.

Key	**Function**
<X]	The <X] (Delete) key sends a DEL character (7/15).
Tab	The **Tab** key sends an HT character (0/9).
Return	The **Return** key sends either a CR character (0/13) or a CR character (0/13) and an LF character (0/10), depending on the set/reset state of line feed/new line mode (LNM). (See Chapter 4).
Ctrl	The **Ctrl** key alone does not send a code. It is always used in combination with another key to send a control code.
Lock	The **Lock** key alone does not send a code. It is used in combination with the "Caps Lock" and "Shift Lock" field selected in the Keyboard Set-Up screen. **Lock** sets or clears the "Caps Lock" (or "Shift Lock") state.
Shift (2 keys)	The **Shift** key alone does not send a code. It is used in combination with other standard keys to generate uppercase characters. **Shift** is also used in combination with nonalphabetic keys to send the top character shown on those keys.
Space bar	The space bar sends an SP character (2/0).
Compose Character	The **Compose Character** key does not send a code. Pressing **Compose Character** starts a compose sequence, which is used to create characters that cannot be typed directly from the keyboard.

3.2.2 Editing Keypad

The editing keypad includes editing keys and cursor control keys. Table 3-1 defines the codes generated by the editing keys, and Table 3-2 defines the codes sent by the cursor control keys.

3.2.3 Auxiliary Keypad

The characters sent by the auxiliary keypad keys depend on the setting of two features, the operating mode (ANSI or VT52) and the keypad (application or numeric) features. The application keypad feature is usually selected only by the computer, but can be selected in the General Set-Up screen. See Chapter 4 for more information about keypad character selection.

Table 3-3 lists the character codes generated by the auxiliary keypad in ANSI (VT100, VT200) mode and in VT52 mode.

3.2.4 Top-Row Function Keys

There are 20 top-row function keys, F1 through F20. The first five keys (F1 through F5) labeled **Hold Screen**, **Print Screen**, **Set-Up**, **Data/Talk**, and **Break**, do not send codes; they are local function keys. Keys **F6** through **F20** send the codes defined in Table 3-4.

3.2.5 Control Codes

Table 3-5 lists the keys and key combinations used to send control codes. These keys and combinations are valid on all keyboards. The control codes are C0 7-bit control characters; there is no similar mechanism for sending C1 8-bit control characters.

3.3 Enabling and Disabling Auto Repeat

You can enable and disable the auto repeat feature from the keyboard, using the Keyboard Set-Up screen or the DECARM escape sequence (Chapter 4). If the terminal receives the DECARM sequence to turn auto repeat off while an auto repeat is in progress, the key stops auto repeating. If the terminal receives the escape sequence to turn auto repeat on while a key that has been auto repeating is still held down, the key immediately auto repeats. Keys which can auto repeat usually start auto repeating after a delay of 0.5 seconds.

The auto repeat speed is a function of the host transmit speed. This gives a constant repeat rate at all transmit speeds. At speeds of 2400 baud or above, all keys auto repeat at 30 keystrokes per second. For the purpose of the auto repeat feature, the keyboard is divided into the following three groups.

```
Group A  Main typing array
Group B  Cursor keys and keypad keys
Group C  Top-row function keys and editing keys
```

Every key in each group auto repeats at the fixed rate set by the transmit speed, regardless of how many codes the key actually sends.

Host Transmit Speed (Baud)	**Auto Repeat Rate (Keys/Sec)**		
	Group A	**Group B**	**Group C**
>=2400	30	30	30
1200	30	30	24
600	30	20	12
300	30	12	12
150	6	6	6
110	6	6	6
75	6	6	6

In general, the "Transmit Rate Limit" feature (in the Communications Set-Up screen) does not affect auto repeat rates, since all 5 codes can be sent at the limited speed of 150 characters per second at most baud rates. In local mode, keys will auto repeat at 30 keystrokes per second.

The following keys do not auto repeat: **Hold Screen**, **Print Screen**, **Set-Up**, **Data/Talk**, **Break**, **Compose Character**, **Shift**, **Return**, **Lock**, and **Ctrl**. Shifted or controlled keys will auto repeat.

3.4 Keyboard Lock and Unlock

The following conditions may cause the keyboard to lock.

- If the program sends a command to set the keyboard action mode (KAM) as described in Chapter 4
- If the keyboard output buffer is full

When the keyboard is locked, all keyboard keys except **Hold Screen**, **Print Screen**, **Set-Up**, **Data/Talk**, and **Break** are disabled, and the keyboard Wait indicator turns on.

If the keyboard is locked, any of the following events can unlock it.

- The output buffer becomes less than full (assuming KAM, keyboard action mode, is not set).

- A KAM reset sequence is received when the output buffer is not full (Chapter 4). Also the DECSTR sequence, and set-up "Reset Terminal" field reset KAM.
- You select "Clear Comm", "Reset Terminal", "Recall", or "Default" fields in set-up. (Entering set-up unlocks the keyboard while the terminal is in set-up. If you do not select these functions in set-up, the keyboard locks again when you exit set-up).
- The terminal performs the power-up self-test (DECTST) or a hard reset (RIS).

4 Received Codes

4.1 General

This chapter describes the terminal's response to codes it may receive from an application or host system. This Chapter assumes you are familiar with the character encoding conventions and terminology covered in Chapter 2.

All data received by the VT220 consists of single and multiple-character codes: graphic (printing or display) characters, control characters, escape sequences, control sequences and device control strings. Much of that data consists of graphic characters that simply appear on the screen with no other effect. Control characters, escape sequences, control sequences and device control strings are all "control functions" that you use in your program to specify how the terminal should process, transmit, and display characters. Each control function has a unique name and each name has a unique abbreviation (mnemonic). Both the name and the abbreviation are standardized.

By default, the terminal interprets individual control and graphic characters according to the DEC Multinational Character Set code mapping (Chapter 2).

NOTE: The terminal usually ignores control codes it does not support. However, codes sent to the terminal other than those specified in this manual can cause unexpected results.

The codes described in this chapter are described as used in VT220 mode unless otherwise indicated.

4.2 Control Characters

Tables 4-1 and 4-2 define the action taken by the terminal when receiving C0 and C1 control characters. The VT220 does not recognize all C0 or C1 characters. Those not shown in either table are ignored (no action taken).

Table 4-1 shows that SO (0/14) and SI (0/15) are also called LS1 and LS0, respectively. SO and SI (shift out and shift in) are the traditional ASCII names or mnemonics. LS1 and LS0 (lock shift G1 and lock shift G0) are alternate names that are useful when dealing with the variety of character set mappings possible. LS1 and LS0 are the names used in this chapter.

Table 4-2 shows the equivalent 7-bit code extension for each 8-bit C1 code. The code extensions require one more byte than the C1 codes. Chapter 2 describes when to use C1 codes and when to use 7-bit code extensions.

4.3 Compatibility Level (DECSCL)

You can set the terminal to a particular level of operation for compatibility with an application. There are two possible levels, level 1 (VT100 operation) and level 2 (VT200 operation). You can set the VT220 (a level 2 terminal) to either level 1 or level 2. These functional levels are defined in Table 4-3.

You can set the compatibility level of the terminal by using the following sequences.

NOTE: You only have to apply restrictions for a lower compatibility level when there is a noncompatible interaction with software written for the lower level.

Sequence	Action
`9/11 3/6 3/1 2/2  7/0` `CSI   6   1   "    p`	Set terminal for level 1 compatibility (VT100 mode).

Sequence	Action
`9/11 3/6 3/2 2/2  7/0` `CSI   6   2   "    p`	Set terminal for level 2 compatibility (VT200 mode, 8-bit controls).

`9/11 3/6 3/2 3/11 3/0 2/2 7/0`
`CSI   6   2   ;    0   "   p`
or
`9/11 3/6 3/2 3/11 3/2 2/2 7/0`
`CSI   6   2   ;    2   "   p`
	Set terminal for level 2 compatibility (VT200 mode, 8-bit controls).

`9/11 3/6 3/2 3/11 3/1 2/2 7/0` `CSI   6   2   ;    1   "   p`	Set terminal for level 2 compatibility (VT200 mode, 7-bit controls).

4.4 Character Set Selection (SCS)

Character encoding in the VT220 was introduced in Chapter 2. The control functions you need to select different graphic character sets are described in this section. Coding differences between the VT220 and VT100 terminals are pointed out where they may affect software compatibility between the VT220 and VT100-type terminals.

The VT220's graphic character repertoire consists of the following graphic sets. (See Chapter 2 for a description of these character sets).

- ASCII graphics
- DEC supplemental graphics (level 2)
- DEC special graphics
- National replacement character (NRC) sets
- Down-line-loadable (level 2)

Generally, you select character sets as follows (Figure 4-1).

NOTE: GR is available only in VT200 mode.

First, you use SCS sequences to designate graphic sets as G0, G1, G2, and G3. This makes the graphics sets available for your program when you map them into GL or GR. To map one of these sets into GL or GR, you must invoke the selected set (G0 through G3) into GL or GR by using locking shifts (LS0, LS1, LS2, LS3, LS1R, LS2R, LS3R) or single shifts (SS2, SS3).

Character sets remain designated until the terminal receives another SCS sequence. All locking shifts remain active until the terminal receives another locking shift. Single shifts SS2 and SS3 are active for only the next single graphic character.

You do not need to select character sets in this manner every time you use the terminal, because there is a default mapping. The default mapping in VT200 mode is ASCII graphics in GL and DEC supplemental graphics in GR (DEC multinational). The default graphic character set mapping is reset whenever you power up the terminal. Your application can also select the default mapping by using the soft terminal reset sequence (DECSTR).

The following sections describe all control functions for designating and invoking graphic character sets.

4.4.1 Designating Hard Character Sets

You designate hard character sets (ASCII, DEC supplemental graphics, DEC special graphics, and national replacement character sets) as G0 through G3, using the following escape sequence formats.

Escape Sequence	Designated As
`1/11 2/8` `ESC ( {final}`	G0
`1/11 2/9` `ESC ) {final}`	G1
`1/11 2/10` `ESC * {final}`	G2
`1/11 2/11` `ESC + {final}`	G3

NOTE: You cannot designate a G2 and G3 set in VT100 mode.

The final character in the escape sequences above represents the character set you want to designate. Table 4-4 lists the available character sets and their associated final character.

To designate a character set as any of the graphic sets (G0 through G3), you must include a final character in one of the escape sequences in Table 4-4. For example, to designate the ASCII character set as the G0 graphic set you would use the following escape sequence.

```
ESC ( B
```

To designate the ASCII character set as the G2 graphic set you would use this escape sequence.

```
ESC * B
```

Note that there is a definite pattern of escape sequences in the designation process.

4.4.2 Designating Soft (Down-Line-Loadable) Character Sets

You can define a soft character set (font) that may or may not replace one of the existing hard sets (ROM fonts). If you do replace a hard set, the replacement occurs for both the 80 and 132-column versions.

A soft character set that replaces a hard character set remains in effect until the soft character set is cleared or redefined. The soft character set is cleared by the "Recall" and "Default" set-up features, and by the power-up self-test. The soft character set is redefined by a DECDLD device control string, described later in this chapter. If the soft character set you define does not replace an existing hard set, then it is used in addition to the hard sets.

NOTE: Only VT220 mode supports the designation of a soft character set.

You designate a soft (down-line-loadable) character set by using the following escape sequences.

Escape Sequence	Designate As
`1/11 2/8 .......` `ESC ( Dscs`	G0
`1/11 2/9 .......` `ESC ) Dscs`	G1
`1/11 2/10 ......` `ESC * Dscs`	G2
`1/11 2/11 ......` `ESC + Dscs`	G3

In these sequences, Dscs is a variable that defines the soft character set used.

Dscs	Function
`I I F`	Generic Dscs. A Dscs can consist of 0 to 2 intermediates (I) and a final (F). Intermediates are in the range 2/0 to 2/15. Finals are in the range 3/0 to 7/14.

Here are three examples of Dscs.

Dscs	Function
`2/0 4/0` `sp   @`	Defines a character set as unregistered soft set. This value is the recommended default value for user-defined sets.
`4/2` `B`	Defines the soft character set to be ASCII.
`2/6 2/5 4/3` `&   %   c`	Defines the character set as "`&%c`", which is currently an unregistered set.

4.4.3 Invoking Character Sets with Locking Shifts

Once you have designated your character sets, you can invoke G0, G1, G2, or G3 into GL or GR. To invoke a set, you use the locking shift (LS) control functions, as shown in Table 4-5 and Figures 4-2 and 4-3.

4.4.4 Invoking Character Sets with Single Shifts

After you designate your character sets, you can invoke G2 or G3 into GL for a single graphic character. To do this, you use the single shift (SS) control functions described below and in Figures 4-2 and 4-3.

All single shifts are active for only the next single graphic character. The terminal returns the previous character set after displaying a single graphic character.

4.4.4.1 SS2 -- Single Shift G2

SS2 is an 8-bit control character (8/14) that invokes G2 into GL for the next graphic character. You can also express SS2 as an escape sequence when coding for a 7-bit environment, as follows.

```
1/11  4/14
ESC    N
```

4.4.4.2 SS3 -- Single Shift G3

SS3 is an 8-bit control character (8/15) that invokes G3 into GL for the next graphic character. You can also express SS3 as an escape sequence when coding for a 7-bit environment as follows.

```
1/11  4/15
ESC    O
```

4.5 Select C1 Controls

You can use *select C1 controls* (code extension announcers) in your program to control the representation of C1 control codes returned to the application by the terminal. (See Chapter 2 for information on working with 7-bit and 8-bit environments.) The terminal always accepts 7-bit or 8-bit forms for C1 controls in either VT200 mode.

Digital recommends that you use DECSCL sequences instead of select C1 controls. The advantage is that DECSCL performs a soft reset, putting the terminal in a known state, in addition to setting the terminal mode and C1 control state.

NOTE: These sequences are supported only in VT200 mode.

4.5.1 Select 7-Bit C1 Control Transmission (S7C1T)

The following sequence converts all C1 codes returned to the application to their equivalent 7-bit code extensions.

```
1/11 2/0  4/7
ESC  sp    F
```

NOTE: The S7C1T sequence is ignored in VT100 or VT52 mode.

4.5.2 Select 8-Bit C1 Control Transmission (S8C1T)

The following sequence returns C1 codes to the application without converting them to their equivalent 7-bit code extensions.

```
1/11 2/0  4/6
ESC  sp    G
```

4.6 Terminal Modes

A *mode* is one of several operating states used by the terminal. Each mode changes the way the terminal works. Table 4-6 lists the selectable modes and their set/reset control sequences. The following sections describe these modes.

Each mode has an identifying name (mnemonic). You can set or reset modes individually or in strings using set mode (SM) or reset mode (RM) control sequences. You can also lock certain features, called user preference features, by using the General Set-Up screen; this prevents the host from changing the feature.

Digital private control sequences (permitted within the extensions of ANSI standards) are identified by DEC in the control sequence mnemonic. These sequences include a question mark character (?) after the control sequence introducer.

You can also select these modes from the keyboard by using set-up screens.

4.6.1 Set Mode (SM)

The set mode sequence for ANSI modes is as follows.

```
9/11      3/11   3/11    6/8
CSI   Ps   ; .... ;   Ps  h
```

The set mode sequence for Digital private modes is as follows.

```
9/11  3/15      3/11   3/11    6/8
CSI    ?    Ps   ; .... ;   Ps  h
```

You use these sequences to set the ANSI and Digital private modes, individually or in strings. Tables 4-7 and 4-8 list the Ps parameters. You cannot use ANSI modes and Digital private modes in the same SM string.

4.6.2 Reset Mode (RM)

The reset mode sequence for ANSI modes is as follows.

```
9/11      3/11   3/11    6/12
CSI   Ps   ; .... ;   Ps  l
```

The reset mode sequence for Digital private modes is as follows.

```
9/11  3/15      3/11   3/11    6/12
CSI    ?    Ps   ; .... ;   Ps  l
```

You use these sequences to reset the ANSI and Digital private modes, individually or in strings. Tables 4-7 and 4-8 list the Ps parameters. You cannot use ANSI modes and Digital private modes in the same RM string.

4.6.3 Keyboard Action Mode (KAM)

Keyboard Action mode lets your program lock and unlock the keyboard. When the keyboard is locked, it cannot send codes to the program. To alert the operator, the terminal turns on the Wait indicators and disables the keyclick features whenever the keyboard is locked.

You can set or reset keyboard action mode as follows.

NOTE: This is a user preference feature that you can lock by using set-up.

Mode	Sequence	Action
Set	`9/11 3/2 6/8` `CSI 2 h`	Locks the keyboard for all following keystrokes.
Reset	`9/11 3/2 6/12` `CSI 2 l`	Unlocks the keyboard, unless it is locked by DC3.

4.6.4 Insert/Replace Mode (IRM)

The terminal displays received characters at the cursor position. Insert/Replace mode determines how the terminal adds characters to the screen. Insert mode displays the new character and moves previously displayed characters to the right. Replace mode adds characters by replacing the character at the cursor position.

You can set or reset insert/replace mode as follows.

Mode	Sequence	Action
Set	`9/11 3/4 6/8` `CSI 4 h`	Selects insert mode. New display characters move old display characters to the right. Characters moved past the right margin are lost.
	`9/11 3/4 6/12` `CSI 4 l`	Selects replacement mode. New display characters replace old display characters at the cursor position. The old character is erased.

4.6.5 Send/Receive Mode (SRM)

Send/receive mode turns local echo on or off. When send/receive mode is reset (local echo on), every character sent from the keyboard automatically appears on the screen. Therefore, the host does not have to send (echo) the character back to the terminal display. When send/receive mode is set (local echo off), the terminal only sends characters to the application. The host must echo the characters back to the screen.

You can set or reset send/receive mode as follows.

Mode	Sequence	Action
Set	`9/11 3/1 3/2 6/8` `CSI 1 2 h`	Turns off (disables) local echo. When the terminal sends characters to the host, the host must echo characters back to the screen.
Reset	`9/11 3/1 3/2 6/12` `CSI 1 2 l`	Turns on (enables) local echo. When the terminal sends characters, the characters are automatically sent to the screen.

4.6.6 Line Feed/New Line Mode (LNM)

Line feed/new line mode selects the control character(s) transmitted to the application by the Return and Enter keys. Enter sends the same code as Return only when the auxiliary keypad is in Keypad numeric mode (DECKPNM).

Line feed/new line also selects the action taken by the terminal when receiving line feed (LF), form feed (FF), or vertical tab (VT) codes. These three codes are always processed identically.

You set and reset line feed/new line mode as follows.

NOTE: For compatibility with Digital software, this mode should always be reset.

Mode	Sequence	Action
Set	`9/11 3/2 3/0 6/8` `CSI 2 0 h`	Causes a received LF, FF, or VT code to move the cursor to the first column of the next line. Return sends both a CR and a LF code.
Reset	`9/11 3/2 3/0 6/12` `CSI 2 0 l`	Causes a received LF, FF, or VT code to move the cursor to the next line in the current column. Return sends a CR code only.

4.6.7 Text Cursor Enable Mode (DECTCEM)

Text cursor enable mode determines if the text cursor is visible.

You set and reset this mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/2 3/5 6/8` `CSI ? 2 5 h`	Makes the cursor visible.
Reset	`9/11 3/15 3/2 3/5 6/12` `CSI ? 2 5 l`	Makes the cursor not visible.

4.6.8 Cursor Key Mode (DECCKM)

Cursor key mode determines the characters sent by the cursor keys.

You can set and reset this mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/1 6/8` `CSI ? 1 h`	Causes the cursor keys to send application control functions.
Reset	`9/11 3/15 3/1 6/12` `CSI ? 1 l`	Causes the cursor keys to generate ANSI cursor control sequences.

4.6.9 ANSI/VT52 Mode (DECANM)

In ANSI mode, reset selects VT52 compatibility mode. In VT52 mode, the terminal responds like a VT52 terminal to private Digital sequences. The reset state of this mode sets the terminal to the VT52 mode. There is no set state for this mode.

```
9/11  3/15  3/2  6/12
CSI    ?     2    l
```

Sets the terminal to VT52 mode.

4.6.10 Column Mode (DECCOLM)

Column mode selects the number of columns per line (80 or 132) on the screen. The screen can display 24 lines of text with either selection.

You select the number of columns per line as follows.

NOTE: When the terminal receives the sequence, the screen is erased and the cursor moves to the home position. This also sets the scrolling region for full screen (24 lines).

Mode	Sequence	Action
Set	`9/11 3/15 3/3 6/8` `CSI   ?    3   h`	Selects 132 columns per line.
Reset	`9/11 3/15 3/3 6/12` `CSI   ?    3   l`	Selects 80 columns per line.

4.6.11 Scrolling Mode (DECSCLM)

Scrolling is the upward or downward movement of existing lines on the screen. There are two methods of scrolling, jump scroll and smooth scroll (6 lines per second).

You can set or reset the scroll mode as follows.

NOTE: This is a user preference feature that you can lock by using set-up.

Mode	Sequence	Action
Set	`9/11 3/15 3/4 6/8` `CSI   ?    4   h`	Selects smooth scroll. Smooth scroll lets the terminal add no more than 6 lines per second to the screen.
Reset	`9/11 3/15 3/4 6/12` `CSI   ?    4   l`	Selects jump scroll. Jump scroll lets the terminal add lines to the screen as fast as possible.

4.6.12 Screen Mode (DECSCNM)

Screen mode selects either a dark or light (reverse) display background on the screen.

You can set or reset screen mode as follows.

NOTE: This is a user preference feature that you can lock by using set-up.

Mode	Sequence	Action
Set	`9/11 3/15 3/5 6/8` `CSI   ?    5   h`	Selects reverse video (dark characters on a light background).

Mode	Sequence	Action
Reset	`9/11 3/15 3/5 6/12` `CSI ? 5 l`	Selects normal screen (light characters on a dark background).

4.6.13 Origin Mode (DECOM)

Origin mode allows cursor addressing relative to a user-defined origin. This mode resets when the terminal is powered up or reset. It does not affect the erase in display (ED) function.

You can set or reset origin mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/6 6/8` `CSI ? 6 h`	Selects the home position with line numbers starting at top margin of the user-defined scrolling region. The cursor cannot move out of the scrolling region.
Reset	`9/11 3/15 3/6 6/12` `CSI ? 6 l`	Selects the home position in the upper-left corner of screen. Line numbers are independent of the scrolling region. Use the CUP sequence to move the cursor out of the scrolling region.

4.6.14 Auto Wrap Mode (DECAWM)

This mode selects where received graphic characters appear when the cursor is at the right margin.

You can set or reset auto wrap mode as follows.

NOTE: Regardless of this selection, the tab character never moves the cursor to the next line.

Mode	Sequence	Action
Set	`9/11 3/15 3/7 6/8` `CSI ? 7 h`	Selects auto wrap. Graphic display characters received when the cursor is at the right margin appear on the next line. The display scrolls up if the cursor is at the end of the scrolling region.
Reset	`9/11 3/15 3/7 6/12` `CSI ? 7 l`	Turns off auto wrap. Graphic display characters received when the cursor is at the right margin replace previously displayed characters.

4.6.15 Auto Repeat (DECARM)

Auto repeat mode selects automatic key repeating. When auto repeat mode is set, a key pressed for more than 0.5 seconds automatically repeats the transmission of the character until the key is released. The following keys never auto repeat: Hold Screen, Print Screen, Set-Up, Data/Talk, Break, Return, Compose Character, Lock, Shift, and Ctrl.

You can set or reset auto repeat mode as follows.

NOTE: This is a user preference feature that you can lock by using set-up.

Mode	Sequence	Action
Set	`9/11 3/15 3/8 6/8` `CSI ? 8 h`	Selects auto repeat mode. A key pressed for more than 0.5 seconds automatically repeats.
Reset	`9/11 3/15 3/8 6/12` `CSI ? 8 l`	Turns off auto repeat. Keys pressed do not automatically repeat.

4.6.16 Print Form Feed Mode (DECPFF)

This mode determines whether or not the terminal sends a print termination character after a screen print. The form feed control character (FF) serves as the print termination character.

You can set or reset print form feed mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/1 3/8 6/8` `CSI ? 1 8 h`	Selects form feed (FF) as print termination character. The terminal sends this character to the printer after each print screen operation.
Reset	`9/11 3/15 3/1 3/8 6/12` `CSI ? 1 8 l`	Selects no termination character. The terminal does not send a form feed (FF) to the printer after each print screen operation.

4.6.17 Print Extent Mode (DECPEX)

Print extent mode selects the full screen or the scrolling region to print during a print screen operation.

You can set or reset this mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/1 3/9 6/8` `CSI ? 1 9 h`	Selects full screen to print during a print screen operation.
Reset	`9/11 3/15 3/1 3/9 6/12` `CSI ? 1 9 l`	Selects scrolling region to print during a print screen operation.

4.6.18 Keypad Mode (DECKPAM/DECKPNM)

The auxiliary keypad generates either numeric characters or control functions. Selecting application or numeric keypad mode determines the type of characters.

You can select the keypad mode as follows.

NOTE: When the terminal is powered up or reset, the terminal selects numeric keypad mode.

Mode	Sequence	Action
Application (DECKPAM)	`1/11 3/13` `ESC =`	Selects application keypad mode. Keypad keys send application control functions.
Numeric (DECKPNM)	`1/11 3/14` `ESC >`	Selects numeric keypad mode. Keypad keys send characters that match the numeric, comma, period, and minus sign keys on main keypad. PF1 through PF4 send control functions.

4.6.19 Character Set Mode (DECNRCM)

Character set mode determines whether the terminal uses national replacement character sets or the DEC multinational character set.

You can select this mode as follows.

Mode	Sequence	Action
Set	`9/11 3/15 3/4 3/2 6/8` `CSI ? 4 2 h`	Selects national mode. The terminal now generates 7-bit characters from the NRC sets.

Mode	Sequence	Action
Reset	`9/11 3/15 3/4 3/2 6/12` `CSI   ?    4   2   l`	Selects multinational mode. The terminal can now generate 8-bit characters from the multinational character set, including 7-bit characters from the ASCII set.

4.7 Cursor Positioning

The cursor indicates the active screen position where the next character will appear (in the absence of auto wrap). A number of operations implicitly affect cursor positioning. In addition, you can control cursor movement using the following sequences.

NOTE: Pn is a variable, ASCII coded, numeric parameter. If you select no parameter or a parameter value of 0, the terminal assumes the parameter equals 1.

Name	Sequence	Action
Cursor Up (CUU)	`9/11      4/1` `CSI  Pn   A`	Moves the cursor up Pn lines in the same column. The cursor stops at the top margin.
Cursor Down (CUD)	`9/11      4/2` `CSI  Pn   B`	Moves the cursor down Pn lines in the same column. The cursor stops at the bottom margin.
Cursor Forward (CUF)	`9/11      4/3` `CSI  Pn   C`	Moves the cursor right Pn columns. The cursor stops at the right margin.
Cursor Backward (CUB)	`9/11      4/4` `CSI  Pn   D`	Moves the cursor left Pn columns. The cursor stops at the left margin.
Cursor Position (CUP)	`9/11      3/11     4/8` `CSI  Pl   ;   Pc  H`	Moves the cursor to line Pl, column Pc. The numbering of the lines and columns depends on the state (set/reset) of origin mode (DECOM).
Horizontal And Vertical Position (HVP)	`9/11      3/11     6/6` `CSI  Pl   ;   Pc  f`	Moves the cursor to line Pl, column Pc. The numbering of the lines and columns depends on the state (set/reset) of origin mode (DECOM). Digital recommends using CUP instead of HVP.
Index (IND)	`1/11 4/4` `ESC   D`	IND is an 8-bit control character (8/4). It can be expressed as an escape sequence for a 7-bit environment. IND moves the cursor down one line in the same column. If the cursor is at the bottom margin, the screen performs a scroll-up.
Reverse Index (RI)	`1/11 4/13` `ESM   M`	RI is an 8-bit control character (8/13). It can be expressed as an escape sequence for a 7-bit environment. RI moves the cursor up one line in the same column. If the cursor is at the top margin, the screen performs a scroll-down.

Name	Sequence	Action
Next Line (NEL)	`1/11 4/5` `ESC   E`	NEL is an 8-bit control character (8/5). It can be expressed as an escape sequence for a 7-bit environment. NEL moves the cursor to the first position on the next line. If the cursor is at the bottom margin, the screen performs a scroll-up.
Save Cursor (DECSC)	`1/11 3/7` `ESC   7`	Saves the following in terminal memory. • cursor position • graphic rendition • character set shift state • state of wrap flag • state of origin mode • state of selective erase
Restore Cursor (DECRC)	`1/11 3/8` `ESC   8`	Restores the states described for (DECSC) above. If none of these characteristics were saved, the cursor moves to home position; origin mode is reset; no character attributes are assigned; and the default character set mapping is established.

4.8 Tab Stops

You select tab stop positions on the horizontal lines of the screen. The cursor advances (tabs) to the next tab stop when the terminal receives a horizontal tab code (HT, 0/9). If there is no next tab, HT moves the cursor to the right margin.

You can set and clear the tab stops as follows.

NOTE: This is a user preference feature that you can lock in set-up.

Name	Sequence	Action
Horizontal Tab Set (HTS)	`1/11 4/8` `ESC   H`	HTS sets a tab stop at the current column. HTS is an 8-bit control character (8/8) that you can also express as an escape sequence when coding for a 7-bit environment.
Tabulation Clear (TBC)	`9/11 6/7` `CSI   g`	Clears a horizontal tab stop at cursor position.
	`9/11 3/0 6/7` `CSI   0   g`	Clears a horizontal tab stop at cursor position.
	`9/11 3/3 6/7` `CSI   3   g`	Clears all horizontal tab stops.

4.9 Character Rendition and Attributes

Character rendition and attributes are display features that affect the way a character is displayed, without changing the character. You can change character rendition using select graphic rendition (SGR) sequences. You can also designate characters to be erasable or not erasable when by using the select character attribute (DECSCA) sequence. DECSCA affects erase sequences.

4.9.1 Select Graphic Rendition (SGR)

You can select one or more character renditions at a time by using the following format.

```
9/11    3/11         6/13
CSI  Ps  ;   Ps ...   m
```

When you use multiple parameters, they are executed in sequence. The effects are cumulative. For example, you can change from increased intensity to blinking-underlined as follows.

```
9/11 3/0 3/11 3/4 3/11 3/5 6/13
CSI   0   ;    4   ;    5   m
```

When you select a single parameter, no delimiter (3/11) is used. For example, you can select the blinking only parameter as follows.

```
9/11 3/5 6/13
CSI   5   m
```

After you select an attribute, all new characters received by the terminal appear with that attribute. If you move the characters by scrolling, the attribute moves with the characters.

You can select character attributes by using the formats described above and the following Ps parameter values.

Ps	Action
3/0 0	All attributes off.
3/1 1	Display bold.
3/4 4	Display underscored.
3/5 5	Display blinking.
3/7 7	Display negative (reverse) image.
3/2 3/2 2 2	Display normal intensity.
3/2 3/4 2 4	Display not underlined.
3/2 3/5 2 5	Display not blinking.
3/2 3/7 2 7	Display positive image.

4.9.2 Select Character Attributes (DECSCA)

You can select all characters to be erasable or not erasable by using the following format. (See the "Erasing" section in this chapter.)

NOTE: This sequence is supported only in VT200 mode.

```
9/11     2/2  7/1
CSI  Ps   "    q
```

Ps is one of the following values.

Ps	Action
0	All attributes off. (Does not apply to SGR.)
1	Designate character as not erasable by DECSEL/DECSED. (Attribute on.)
2	Designate character as erasable by DECSEL/DECSED. (Attribute off.)

NOTE: A parameter value of 0 implies the default, which is attributes off (erasable by DECSEL/DECSED). A parameter value of 2 is an explicit request for this attribute to be off (so the character is erasable by DECSEL/DECSED).

4.10 Line Attributes

Lines attributes are display features that affect a complete display line. The cursor selects the line affected by the attribute. The cursor stays in the same character position when the attribute changes, unless the attribute would move the cursor past the right margin. In that case, the cursor stops at the right margin. When you move lines on the screen by scrolling, the attribute moves with the line. Select line attributes by using the following sequences.

NOTE: If you erase an entire line while using the erase in display (ED) sequence, the line attribute changes to single-height and single-width.

4.10.1 Double Height Line (DECDHL)

These sequences make the line with the cursor the top or bottom half of a double-height, double-width line. You must use these sequences in pairs on adjacent lines. The same character must be used on both lines to form a full character. If the line was previously single-width, single-height, all characters to the right of center are lost.

Top Half	Bottom Half
1/11 2/3 3/3 ESC # 3	1/11 2/3 3/4 ESC # 4

4.10.2 Single-Width Line (DECSWL)

The DECSWL sequence makes the line with the cursor single-width, single-height. This is the line attribute for all new lines on the screen.

```
1/11 2/3 3/5
ESC   #   5
```

4.10.3 Double-Width Line (DECDWL)

The DECDWL sequence makes the line with the cursor double-width, single-height. If the line was previously single-width, single-height, all characters to the right of center screen are lost.

```
1/11 2/3 3/6
ESC   #   6
```

4.11 Editing

You use editing sequences to insert and delete characters and lines of characters at the cursor position. The cursor position does not change when inserting or deleting lines.

You can delete characters or insert and delete lines as follows.

NOTE: Pn is a variable, ASCII-coded, numeric parameter. If you select no parameter or a parameter value of 0, the terminal assumes a parameter value of 1.

Name	Sequence	Action
Insert Line (IL)	`9/11 4/12` `CSI Pn L`	Inserts Pn lines at the cursor. If fewer than Pn lines remain from the current line to the end of the scrolling region, the number of lines inserted is the lesser number. Lines within the scrolling region at and below the cursor move down. Lines moved past the bottom margin are lost. The cursor is reset to the first column. This sequence is ignored when the cursor is outside the scrolling region.
Delete Line (DL)	`9/11 4/13` `CSI Pn M`	Deletes Pn lines starting at the line with the cursor. If fewer than Pn lines remain from the current line to the end of the scrolling region, the number of lines deleted is the lesser number. As lines are deleted, lines within the scrolling region and below the cursor move up, and blank lines are added at the bottom of the scrolling region. The cursor is reset to the first column. This sequence is ignored when the cursor is outside the scrolling region.
Insert Characters (ICH) (VT200 mode only)	`9/11 4/0` `CSI Pn @`	Insert Pn blank characters at the cursor position, with the character attributes set to normal. The cursor does not move and remains at the beginning of the inserted blank characters. A parameter of 0 or 1 inserts one blank character. Data on the line is shifted forward as in character insertion.
Delete Character (DCH)	`9/11 5/0` `CSI Pn P`	Deletes Pn characters starting with the character at the cursor position. When a character is deleted, all characters to the right of the cursor move to the left. This creates a space character at the right margin for each character deleted. Character attributes move with the characters. The spaces created at the end of the line have all their character attributes off.

4.12 Erasing

Erasing removes characters from the screen without affecting other characters on the screen. Erased characters are lost. The cursor position does not change when erasing characters or lines. Erasing a character also erases any character attribute of the character.

You can erase characters as follows.

Name	Sequence	Action
Erase Character (ECH) (VT200 mode only)	`9/11 5/8` `CSI Pn X`	Erases characters at the cursor position and the next Pn-1 characters. A parameter of 0 or 1 erases a single character. Character attributes are set to normal. No reformatting of data on the line occurs. The cursor remains in the same position.
Erase in Line (EL)	`9/11 4/11` `CSI K`	Erases from the cursor to the end of the line, including the cursor position. Line attribute is not affected.
	`9/11 3/0 4/11` `CSI 0 K`	Same as above.
	`9/11 3/1 4/11` `CSI 1 K`	Erases from the beginning of the line to the cursor, including the cursor position. Line attribute is not affected.
	`9/11 3/2 4/11` `CSI 2 K`	Erases the complete line.
Erase in Display (ED)	`9/11 4/10` `CSI J`	Erases from the cursor to the end of the screen, including the cursor position. Line attribute becomes single-height, single-width for all completely erased lines.
	`9/11 3/0 4/10` `CSI 0 J`	Same as above.
	`9/11 3/1 4/10` `CSI 1 J`	Erases from the beginning of the screen to the cursor, including the cursor position. Line attribute becomes single-height, single-width for all completely erased lines.
	`9/11 3/2 4/10` `CSI 2 J`	Erases the complete display. All lines are erased and changed to single-width. The cursor does not move.
Selective Erase In Line (DECSEL) (VT200 move only)	`9/11 3/15 4/11` `CSI ? K`	Erases all erasable characters (DECSCA) from the cursor to the end of the line. Does not affect video line attributes or video character attributes (SGR).
	`9/11 3/15 3/0 4/11` `CSI ? 0 K`	Same as above.
	`9/11 3/15 3/1 4/11` `CSI ? 1 K`	Erases all erasable characters (DECSCA) from the beginning of the line to and including the cursor position. Does not affect video line attributes or video character attributes.
	`9/11 3/15 3/2 4/11` `CSI ? 2 K`	Erases all erasable characters (DECSCA) on the line. Does not affect video line attributes or video character attributes.

Name	Sequence	Action
Selective Erase In Display (DECSED) (VT200 mode only)	9/11 3/15 4/10 CSI ? J	Erases all erasable characters (DECSCA) from and including the cursor to the end of the screen. Does not affect video line attributes or video character attributes (SGR).
	9/11 3/15 3/0 4/10 CSI ? 0 J	Same as above.
	9/11 3/15 3/1 4/10 CSI ? 1 J	Erases all erasable characters (DECSCA) from the beginning of the screen to and including the cursor. Does not affect video line attributes or video character attributes (SGR).
	9/11 3/15 3/2 4/10 CSI ? 2 J	Erases all erasable characters (DECSCA) in the entire display. Does not affect video character attributes or video line attributes (SGR).

4.13 Scrolling Margins (Top and Bottom)

The scrolling region is the area of the screen that can receive new characters by scrolling old characters off the screen. The area is defined by the top and bottom screen margins. The smallest scrolling region allowed is two lines; therefore the number of the top margin must be at least one less than the number of the bottom margin.

You can select the top and bottom margins of the scrolling region as follows.

4.13.1 Set Top and Bottom Margins (DECSTBM)

This sequence selects top and bottom margins defining the scrolling region.

```
9/11    3/11     7/2
CSI  Pt  ;   Pb   r
```

4.14 Printing

You can select all print operations with control sequences. When characters are printed on the screen, terminal and printer tab stops are ignored. Print characters are spaced with the space (SP) character. The terminal transmits a carriage return (CR) and line feed (LF), a vertical tab (VT), or a form feed (FF) after the last printable character of a line (not a space character).

NOTE: Spaces with video attributes are printable characters.

Before you select a print operation, you should check printer status with the print status report (DSR) (see the "Reports" section in this chapter.)

You can select print operations as follows.

Name	Sequence	Action
Auto Print Mode	9/11 3/15 3/5 6/9 CSI ? 5 i	Turns on auto print mode. Display lines print when you move the cursor off the line with a line feed, form feed, vertical tab, or auto wrap. The printed line ends with a carriage return and the character that moved the cursor off the previous line (LF, FF, or VT). Auto wrap lines end with a linefeed.
	9/11 3/15 3/4 6/9 CSI ? 4 i	Turns off auto print mode.

Name	Sequence	Action
Printer Controller	`9/11 3/5 6/9` `CSI 5 i`	Turns on printer controller mode. The terminal sends received characters to the printer without displaying them on the screen. All characters and character sequences except NUL, XON, XOFF, CSI 5 i, and CSI 4 i are sent to the printer. The terminal does not insert or delete spaces, provide line delimiters, or select the correct printer character set. Printer controller mode has a higher priority than auto print mode. You can select printer controller mode during auto print mode. In printer controller mode, keyboard activity continues to be directed to the host.
	`9/11 3/4 6/9` `CSI 4 i`	Turns off printer controller mode.
Print Cursor Line	`9/11 3/15 3/1 6/9` `CSI ? 1 i`	Prints the display line containing the cursor. The cursor position does not change. The print-cursor-line sequence is complete when the line prints.
Print Screen	`9/11 6/9` `CSI i`	Prints the screen display (full screen or scrolling region, depending on the print extent DECPEX selection). Printer form feed mode (DECPFF) selects either a form feed (FF) or nothing as the print terminator. The print screen sequence is complete when the screen prints.
	`9/11 3/0 6/9` `CSI 0 i`	Same as above.

4.15 User Defined Keys (DECUDK)

Fifteen of the terminal's top-row function keys are programmable: F6 through F14, Do, Help, and F17 through F20. (Hold Screen, Print Screen, Set-Up, Data/Talk and Break have dedicated local functions and are not programmable). When the terminal is in VT200 mode, you can down-line-load one or more key sequences for the programmable function keys by using DECUDK device control strings. (The programmable function keys are inoperative in the VT100 and VT52 modes.)

To access the programmed values of the keys, you type Shift-(function key). To access the normal control sequence values, you press the function key alone.

There are 256 bytes available to the 15 programmable function keys. Space is supplied on a first-come/first-serve basis. Once the 256 bytes are used, you cannot redefine any more keys unless you clear space. There are three ways you can clear space.

1. Redefine a key or keys by using a DECUDK.
2. Clear a key or keys by using a DECUDK.
3. Clear the definition set with a terminal power-up or recall operation.

NOTE: All keys definitions are stored in volatile RAM. UDK key definitions are lost when terminal power is lost.

4.15.1 DECUDK Device Control String Format

NOTE: See Chapter 2 for general information about device control strings.

The device control string format for down-line-loading UDK functions is as follows.

`DCS`	`Pc;Pl`	`|`	`Ky1/st1;ky2/st2;...kyn/stn`	`ST`
Control String Introducer	Clear and Lock Parameters	Final Character	Key Definition String	String Terminator

Each string component is described below.

NOTE: This sequence is supported only in VT200 mode.

The **device control string introducer** is DCS (9/0). This character introduces the control string. DCS is an 8-bit character (9/0) that you can also express as ESC P (1/11 5/0) when coding for a 7-bit environment.

The **clear parameter** (Pc) determines which keys are cleared and when. A value of 0 (clear all) clears all keys, then loads each specific key as it is met in the DRCS. A value of 1 (load new values, clear old only when redefined) clears each key to be reloaded just before loading it; a 1 does not clear keys that are not being redefined. By using a value of 1 for Pc, you can redefine some keys without reloading them all.

NOTE: There are only 256 bytes available. A key definition cannot contain more than 256 bytes or the number of bytes available when that key is loaded, whichever is less.

Note that if you set Pc to 1 (load new, but do not clear old), a key load may fail because no room is available – even though the final total for all keys would have been 256 bytes or less. The reason for this is as follows.

With Pc set to 1, keys are cleared and loaded sequentially. Sequential loading may result in intermediate storage requirements higher than 256 bytes, even though the final requirement would be 256 bytes or less. For example, suppose F6 contained 120 bytes, F7 contained 110 bytes, and F8 contained 20 bytes. Loading F8 with 40 bytes, F6 with 1 byte and F7 with 1 byte works if all keys are cleared first, but not if the keys are cleared as they are sequentially redefined. When you try to load F8 with 40 bytes, the load fails because only 26 bytes are free at that time (256 - 120 - 110 = 26).

The following is a summary of Pc values and meanings.

Pc	Meaning
None	Clear all keys before loading new values.
0	Clear all keys before loading new values.
1	Load new key values, clear old only where defined.

The **lock parameter** (Pl) determines whether the key definitions are locked or not locked after you load them. Pl follows the Pc values and is separated by a semicolon character (`;`, 3/11) as a delimiter. If you set the Pl value to 0 (lock), the keys are locked at the completion of loading. At this point, the terminal operator must unlock the keys for redefinition using set-up. If you set the Pl value to 1 (do not lock), the keys are available for further definition with another DECUDK string. The default for the lock parameter is lock.

NOTE: A Pl value of 1 does not unlock the keys. It simply does not lock them.

The following is a summary of Pl values and meanings.

Pl	Meaning
None	Lock the keys. (Cannot be redefined.)
0	Lock the keys. (Cannot be redefined.)
1	Do not lock the keys. (Can be redefined.)

The **final character**, a vertical bar (7/12), designates this control string as a DECUDK.

The **key definition strings** (Kyn/Stn) are included in the data following the final character and before the string terminator. Each key definition string consists of a key selector number (Kyn) and a string parameter (Stn) separated by a slash (`/`, 2/15).

The key selector numbers (Kyn) specify the particular key to be redefined, and the string parameters (Stn) are the encoded contents of the keys. The string parameters (Stn) consist of hex pairs in the range of 3/0 through 3/9 (0 through 9), 4/1 through 4/6 (A through F), and 6/1 through 6/6 (a through f). When you combine these hex values, they represent an 8-bit quantity.

This method lets you use any of the 256 character codes in the key sequence. You can use key definition strings in any order. You can also specify multiple definitions using a semicolon (`;` 3/11) as a delimiter.

Here is a list of definable keys and their identifying values.

Key	Value
F6	17
F7	18
F8	19
F9	20
F10	21
F11	23
F12	24
F13	25
F14	26
Help	28
Do	29
F17	31
F18	32
F19	33
F20	34

The **string terminator** is `ST` (9/12). This is an 8-bit control character that you can also express as `ESC \` (1/11, 5/12) when coding for a 7-bit environment.

NOTE: To access the programmed values of keys, you type Shift-(function key). To access the normal control sequences, you type the function key alone.

4.15.2 Things to Keep in Mind When Loading Keys

Here are some general guidelines you should keep in mind when loading the keys.

- Software should use the UDK function to clear key definition space. You can do this by clearing keys without locking them. After the keys are cleared, you can use the UDK function to redefine the keys and lock them.
- Generally, you should not leave keys unlocked. This could cause a breach of security for the terminal user and the computer system.

- The host must keep track of space available for definitions.
- If you redefine a key, the old sequence is lost. This may clear some space if the new sequence is shorter than the previous definition.
- The terminal uses a special lock to arbitrate the programming of keys. You can turn this lock on or off in set-up. It may also be turned on with a DECUDK from the host. The lock acts globally over all programmable keys.
- The default value for each key definition is empty (blank). When you clear the keys, they are empty. All key definitions are stored in volatile RAM. If the terminal loses power, key definitions are lost. An aborted function key load (by error or other cause) locks the keys, saves the already successfully loaded fraction, and sends the rest of the DECUDK sequence to the screen. An invalid DCS hex pair in a key definition string causes an aborted load.

4.15.3 Examples and Recommendations for Using DECUDK

To clear keys, send the following sequence.

```
9/0  3/0  3/11  3/1  7/12  9/12
DCS   0    ;     1    |     ST
```

To lock keys, send the following sequence.

```
9/0  3/1  3/11  3/0  7/12  9/12
DCS   1    ;     0    |     ST
```

Suppose you want to define key F20 to be "PRINT", and you want to do this without clearing or locking any other keys. The first part of your sequence would look like this.

```
9/0  3/1  3/11  3/1  7/12  3/3  3/4  2/15
DCS   1    ;     1    |     3    4    /
```

The 34 after the final character (7/12) identifies key F20. After the slash character (2/15), you would include the definition. The encoding for "PRINT" is as follows.

```
P = 50 hex
R = 52 hex
I = 49 hex
N = 4E hex
T = 54 hex
```

So the rest of the sequence you enter after the slash character would look like this.

```
3/5  3/0  3/5  3/2  3/4  3/9  3/4  4/5  3/5  3/4  9/12
 5    0    5    2    4    9    4    E    5    4    ST
```

The `ST` character (9/12) marks the end of the sequence.

4.16 Down-Line-Loadable Character Set

In a VT200 mode, the terminal lets you create and down-line-load a character set containing up to 94 characters. This character set is called a dynamically redefinable character set (DRCS). After you create characters, you can load them into the terminal DRCS buffer by using a DECDLD device control string.

NOTE: This character set is not loaded into nonvolatile RAM. So when the terminal is powered off, characters are lost.

4.16.1 Designing a Character Set

Figure 4-4 compares DRCS cell size and the terminal's normal character cell size in pixels. Note that the maximum number of pixels in the DRCS cell is 80 (8 × 10). Note also, that the normal terminal's character cell size is less than the DRCS cell size. Since the terminal ignores characters defined beyond the DRCS cell size, you must design your characters to fit the DRCS cell size.

Each pixel in a character font is represented by a bit with a binary value of 1 (on) or 0 (off). One (1) specifies foreground (pixel on) and zero (0) specifies background (pixel off).

For example, suppose you want to design character A. To do this, you follow a four-step process.

1. First designate which pixels are to be on and which pixels will be off. Your design may look like Figure 4-5.
2. After you establish what your DRCS character A will look like, divide the pixels of the DRCS character cell into columns of six bits each using the format shown in Figure 4-6. The column numbers here designate the order in which the columns are sent to the terminal.

   Each column is now represented as a vertical 1 × 6 pixel matrix called a *sixel*. The least significant bit is at the top, and the most significant bit is at the bottom. Because the character height (10 pixels) is not a multiple of 6, the columns on the bottom of the character cell have only 4 bits each. (The two highest order bits, 5 and 6, are ignored.)
3. After you divide your DRCS character into six-pixel columns (sixels), convert the binary values of each column to its equivalent character. Because column codes are restricted to characters with the range of `?` (octal 077) to `~` (octal 176), you must add an offset of octal 077 to each column octal value. Thus, binary value 000000 is converted to octal 077 (octal 0 + octal 77); binary 110101 is converted to octal 164 (octal 65 + octal 077); and binary 111111 is converted to octal 176 (octal 077 + 077).
4. After you convert the binary column codes to octal values (using the offset), convert the octal value for each column to its equivalent character using the ASCII table in Chapter 2. Figure 4-7 provides this conversion procedure for our example of a DRCS character A.

Use this procedure to design each DRCS characters you want. You can then down-line-load your DRCS characters with the DECDLD device control string described in the next section. The DRCS characters consist of a string or strings of characters.

4.16.2 Down-Line-Loading DRCS Characters

You can down-line-load your DRCS character set with the following DECDLD device control string format.

NOTE: See Chapter 2 for general information about device control strings.

```
DCS Pfn;Pcn;Pe;Pcms;Pw;Pt { Dscs Sxbp1;Sxbp2;...;Sxbpn ST
```

`DCS` (9/0) is the device control string introducer. It is an 8-bit control character that you can also express as `ESC P` (1/11, 5/0) when coding for a 7-bit environment.

`Pfn;Pcn;Pe;Pcms;Pw;Pt` are parameter characters, separated by semicolons. Table 4-9 describes these parameters. Valid values for Pcms, Pw, and Pt are: Pcms = 0, 2, 3 or 4; Pw = 0, 1 or 2; and Pt = 0 or 1. Invalid combinations are ignored. You can use a loaded font (character set) in 80 or 132 columns.

`{` (7/11) is the final character that marks the end of the parameter characters and specifies a DECDLD function.

Dscs defines the character set name for the soft font, and is used in the SCS (select character set) escape sequence.

Sxbp1;Sxbp2;...;Sxbpn are sixel bit patterns (1 to 94 patterns) for characters separated by semicolons. Each sixel bit pattern has the form:

```
S...S/S...S
```

where

- the first S...S represents the upper columns (sixels) of the DRCS character,
- the slash (2/5) advances the sixel pattern to the lower columns of the DRCS character,
- and the second S...S represents the lower columns (sixels) of the DRCS (Figure 4-6).

ST (9/12) is the string terminator. It is an 8-bit control character that you can also express as `ESC \` (1/11, 5/12) when coding for a 7-bit environment.

4.16.3 DECDLD Example

Suppose you want to load a character set starting with the character A, designed in the example in Paragraph 4.16.1. To do this, you could use the following device control string.

```
DCS 1;1;1 { sp @ ogcacgo/B?????B;(next character);.....ST
```

DCS introduces the sequence.

1;1;1 specifies loading the DRCS font buffer, selects the starting character as column 2/row 1 of ASCII chart (Chapter 2), and selects to erase only the characters that are loaded (Table 4-9). Note that Pcms, Pw, and Pt are not included; they default to 0 values.

`{` indicates the end of the parameter characters and specifies that this is a DECDLD control string.

sp @ defines a character set as an unregistered soft set. This value is the recommended default value for user-defined sets. The "sp" represents one space. Other DRCS values can be used to define other specific character sets.

ogcacgo are the character codes for the upper columns of the example DRCS character A.

`/` advances the sixel sequence to the lower columns of the example DRCS character A.

B?????B are the character codes for the lower columns of the example DRCS character A.

; signals the end of the DRCS character string being loaded and the beginning of another DRCS character to be loaded.

ST indicates the end of the device control string.

NOTE: See the "Character Set Selection" section in this chapter for information on designating and invoking soft character sets.

4.16.4 Clearing a Down-Line-Loaded Character Set

You can clear a character set that you have down-line-loaded by using the following DECDLD control sequence:

```
DCS 1;1;2 { sp @ ST
```

Down-line-loaded character sets are also cleared by the following actions.

- Performing the power-up self test
- Using the set-up "Recall" or "Default" fields
- Using RIS or ESC c sequences

4.17 Reports

The terminal sends reports in response to host computer requests. These reports provide identification (type of terminal), cursor position, and terminal operating status. There are two categories of reports, device attributes (DA) and device status reports (DSR).

4.17.1 Device Attributes (DA)

There are two DA exchanges (dialogues) between the host computer and the VT220, primary DA and secondary DA.

4.17.1.1 Primary DA

In the primary DA exchange (the first exchange), the host asks for the terminal's service class code and the basic attributes. The terminal's response depends on the value of the "Terminal ID" field selected in set-up.

A typical primary DA exchange is as follows.

Communication	Sequence	Meaning
Host to VT220	`CSI c` or `CSI 0 c`	"What is your service class code and what are your attributes?"
VT220 to host	`CSI ? 62; 1; 2; 6; 7; 8; 9 c`	"I am a service class 2 terminal (62) with 132 columns (1), printer port (2), selective erase (6), DRCS (7), UDK (8), and I support 7-bit national replacement character sets (9)."

NOTE: If the terminal is in VT100 mode and you select an ID other than VT220 ID, then the following primary exchanges apply.

VT220 to host (VT100 ID selected in set-up)	`ESC [ ? 1; 2 c`	"I am a VT100 terminal with AVO."
VT220 to host (VT101 ID selected in set-up)	`ESC [ ? 1; 0 c`	"I am a VT101 terminal."
VT220 to host (VT102 ID selected in set-up)	`ESC [ ? 6 c`	"I am a VT102 terminal."

4.17.1.2 Secondary DA

In the secondary DA exchange (the second exchange), the host asks for the terminal's identification code, firmware version level, and an account of the hardware options installed.

A typical secondary DA is as follows.

Communication	Sequence	Meaning
Host to VT220 (Secondary DA request)	`CSI > c` or `CSI > 0 c`	"What type of terminal are you, what is your firmware version, and what hardware options do you have installed?"
VT220 to Host (Secondary DA response)	`CSI > 1; Pv; Po c`	"I am a VT220 (identification code of 1), my firmware version is _____ (Pv), and I have _____ Po options installed."

Example

`CSI>1;10;0c` = VT220 version 1.0, no options.

4.17.2 Device Status Report (DSR)

In a DSR exchange, the host computer asks for the general operating status of the terminal and/or printer. If the terminal is in print controller mode, the printer receives the DSR request but cannot answer.

DSR – VT220

A typical DSR exchange is as follows.

Communication	Sequence	Meaning
Host to VT220 (Request for terminal status)	`CSI 5 n`	"Please report your operating status using a DSR control sequence. Are you in good operating condition or do you have a malfunction?"
VT220 to host (DSR Response)	`CSI 0 n`	"I have no malfunction."
	or	
	`CSI 3 n`	"I have a malfunction."
Host to VT220 (Request for cursor position)	`CSI 6 n`	"Please report your cursor position using a CPR (not DSR) control sequence."
VT220 to host (CPR response)	`CSI Pv; Ph R`	"My cursor is positioned at _____ (Pv); _____ (Ph)." Pv = vertical position (row) Ph = horizontal position (column)

DSR – Printer Port

NOTE: Determine the printer status before entering any print mode or starting any print function.

A typical DSR exchange is as follows.

Communication	Sequence	Meaning
Host to VT220 (Request for printer status)	`CSI ? 15 n`	"What is the printer status?"
VT220 to host	`CSI ? 13 n`	"DTR has not been asserted on the printer port since power up or reset – in effect, I have no printer."
	`CSI ? 10 n`	"DTR is asserted on the printer port. The printer is ready."
	`CSI ? 11 n`	"DTR is not currently asserted on the printer port. The printer is not ready."

DSR – User Defined Keys (VT220 mode only)

A typical DSR exchange is as follows.

Communication	Sequence	Meaning
Host to VT220 (Request for UDK status)	`CSI ? 25 n`	"Are user-defined keys locked or unlocked?"
VT220 to host	`CSI ? 20 n`	"User-defined keys are unlocked."
	`CSI ? 21 n`	"User-defined keys are locked."

DSR – Keyboard Language

A typical DSR exchange is as follows.

Communication	Sequence	Meaning
Host to VT220 (Request for keyboard language)	`CSI ? 26 n`	"What is the keyboard language?"
VT220 to host	`CSI ? 27; Pn n`	"My keyboard language is _____ (Pn)."

where:

Pn		Language
0	=	Unknown*
1	=	North American
2	=	British
3	=	Flemish
4	=	French Canadian
5	=	Danish
6	=	Finnish
7	=	German
8	=	Dutch
9	=	Italian
10	=	Swiss (French)
11	=	Swiss (German)
12	=	Swedish
13	=	Norwegian
14	=	French/Belgian
15	=	Spanish

* Sent by a terminal that for some reason cannot determine its keyboard language. The VT220 will never send this response.

4.17.3 Identification (DECID)

The DECID sequence causes the terminal to send a primary DA response sequence. Digital does not recommend using DECID. You should use the primary DA request for this purpose.

The DECID sequence is as follows.

```
1/11 5/10
ESC   Z
```

4.18 Terminal Reset (DECSTR and RIS)

There are two terminal reset escape sequences. One causes a soft terminal reset (DECSTR), and the other causes a hard terminal reset (RIS).

4.18.1 Soft Terminal Reset (DECSTR)

You can invoke DECSTR (soft terminal reset) from the keyboard by selecting "Reset Terminal" in the Set-Up Directory screen. It can be invoked directly from the host computer via the DECSTR sequence, if the terminal is in a VT200 mode. (When the terminal is in VT100 or VT52 mode, the escape sequence is ignored.) It can also be invoked indirectly via the DECSCL sequence (ignored in VT52 mode).

The DECSTR sequence sets the terminal to the power-up default states listed in Table 4-10.

The DECSTR escape sequence is as follows.

```
9/11 2/1 7/0
CSI   !   p
```

4.18.2 Hard Terminal Reset (RIS)

You can invoke RIS (hard terminal reset or reset to initial state) from the keyboard by selecting "Recall" in the Set-Up Directory screen. It can also be invoked from the host computer with an escape sequence. RIS causes an NVR recall. All set-up parameters are replaced either by their NVR values, or power-up default values if NVR values do not exist.

In addition, RIS performs the following actions.

- Performs a communications line disconnect.
- Clears UDKs.
- Clears a down-line-loaded character set.
- Clears the screen.
- Returns the cursor to the upper-left corner of the screen.
- Sets the SGR state to normal.
- Sets the selective erase attribute write state to "not erasable".
- Sets all character sets to the default.

The RIS escape sequence is as follows.

```
1/11 6/3
ESC   c
```

NOTE: Use this sequence with caution. Parity and baud rate are restored from NVR.

4.19 Tests and Adjustments (DECTST and DECALN)

The terminal has tests and alignment patterns you can invoke from the keyboard or from the host computer with control and escape sequences. Test and alignment procedures are usually performed only by Digital Manufacturing and Field Service personnel.

This section provides the sequences used to invoke the tests and the alignment patterns. For detailed information, see the *VT220 Pocket Service Guide*.

4.19.1 Tests (DECTST)

The sequence format for invoking terminal tests is as follows.

```
9/11 3/4  3/11     3/11    3/11     7/9
CSI   4    ;   Ps   ; ..... ;   Ps   y
```

Each Ps is a parameter indicating a test to perform. After the first parameter (4), the parameters each select one test from the following list. You can invoke several tests at one time by separating the parameters with semicolons. The tests are not necessarily executed in the order you enter them.

NOTE: DECTST causes a communications line disconnect.

Parameter	Test
0	Test 1, 2, 3 and 6
1	Power-up self test
2	EIA port data loopback test
3	Printer port loopback test
4	Not used
5	Not used
6	EIA port modem control line loopback test
7	20 mA port loopback test
8	Not used
9	Repeat any selected test continuously until power-off or failure
10 and up	Not used

4.19.2 Adjustments (DECALN)

The terminal has a screen alignment pattern that service personnel use to adjust the screen. You can display the screen alignment pattern with the DECALN sequence.

```
1/11 2/3 3/8
ESC   #   8
```

This sequence fills the screen with uppercase E's.

4.20 VT52 Mode Escape Sequences

The VT52 mode allows the VT220 to operate with Digital software written for the VT52 terminal. In VT52 mode, all C0 control characters are allowed, although some are ignored. No C1 control characters or ANSI mode control functions are allowed. The user-defined keys are disabled. Table 4-11 defines the VT52 mode escape sequences. Table 3-3 in Chapter 3 defines the VT52 auxiliary keypad codes. The set-up VT100 (ASCII/UK) default character set applies to VT52 mode as well.

VT220/VT102 Differences

This appendix describes the major differences between a VT102 terminal and the VT220 terminal operating in a VT100 mode.

Difference Area	**VT102**	**VT220**
LED indicators	Programmable	Not Programmable
Alternate character ROM	Socket for OEM-supplied character ROM	Down-line-loadable character set
Screen freeze	NO SCROLL key	Hold Screen key
Printer port connector	25-pin	9-pin
Screen refresh	50/60 Hz	60 Hz only
Set-up	Display in set-up "bits".	Display in three different languages. Different display. Has no effect on software.
RIS function	Performs power-up self test	Does not perform power-up self test
Off-Line/local mode	Off-Line mode disconnects modem.	Local mode does not disconnect modem.
Communication	Full duplex and half duplex	Full duplex only. Does not affect software.
	Transmit speed limit is 60 characters/second regardless of baud rate.	Optional transmit speed limit is 150 characters/second.
	Passive or active 20 mA is selectable.	Only has passive 20 mA.
Terminal ID	Responds to primary device attributes with `ESC [ ? 6 c` sequence.	You can select the DA response in set-up. The standard VT220 response is `CSI ? 62;1;2;6;7;8;9 c` sequence.
	Secondary device attribute not supported.	Responds to secondary device attribute with `CSI > 1;Pv; 0 c` sequence. Pv is firmware version number.

Additional VT220 Documentation

This appendix lists and describes additional VT220 documents that can be ordered from DIGITAL.

Title	Number	Description
VT220 Owner's Manual	EK-VT220-UG	The VT220 Owner's Manual provides the information needed to operate and maintain the VT220.
VT220 Installation Guide	EK-VT220-IN	The VT220 Installation Guide describes the installation procedure for the VT220.
VT220 Programmer Pocket Guide	EK-VT220-HR	The VT220 Programmer Pocket Guide provides a quick-reference summary of VT220 programming information.
VT220 Pocket Service Guide	EK-VT220-PS	The VT220 Pocket Service Guide provides procedures for troubleshooting and repairing the VT220 to the field replaceable unit.
VT220 Illustrated Parts Breakdown (IPB)	EK-VT220-IP	The VT220 IPB provides an illustrated parts breakdown of the VT220.
VT220 Family Field Maintenance Print Set	MP-01732-01	The VT220 Family Field Maintenance Print Set provides a complete set of VT220 electrical and mechanical schematic diagrams.

Tables

Table 2-1: 7-Bit ASCII Code Table

ROW	COLUMN				0		1		2		3		4		5		6		7	
	BITS b7 b6 b5				0 0 0		0 0 1		0 1 0		0 1 1		1 0 0		1 0 1		1 1 0		1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	[	133 91 5B	k	153 107 6B	{	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	\	134 92 5C	l	154 108 6C	|	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	]	135 93 5D	m	155 109 6D	}	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	~	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-2 8-Bit Code Table

COLUMN / ROW	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
00	NUL	DLE	SP							DCS	///					
01	SOH	DC1								PU1						
02	STX	DC2								PU2						
03	ETX	DC3								STS						
04	EOT	DC4							IND	CCH						
05	ENQ	NAK							NEL	MW						
06	ACK	SYN							SSA	SPA						
07	BEL	ETB							ESA	EPA						
08	BS	CAN							HTS							
09	HT	EM							HTJ							
10	LF	SUB							VTS							
11	VT	ESC							PLD	CSI						
12	FF	FS							PLU	ST						
13	CR	GS							RI	OSC						
14	SO	RS							SS2	PM						
15	SI	US						DEL	SS3	APC						///

C0 CODES (columns 00–01) · GL CODES (columns 02–07) · C1 CODES (columns 08–09) · GR CODES (columns 10–15)

7-BIT CODE TABLE (columns 00–07)

Table 2-3: DEC Multinational Character Set (C0 and GL Codes) (Sheet 1 | 2)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	[	133 91 5B	k	153 107 6B	{	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	\	134 92 5C	l	154 108 6C	|	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	]	135 93 5D	m	155 109 6D	}	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	~	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F
					C0 CODES				GL CODES (ASCII GRAPHICS)											

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-3: DEC Multinational Character Set (C1 and GR Codes) (Sheet 1 | 2)

8		9		10		11		12		13		14		15		COLUMN				ROW
1 0 0 0		1 0 0 1		1 0 1 0		1 0 1 1		1 1 0 0		1 1 0 1		1 1 1 0		1 1 1 1		b8 b4	b7 b3	b6 b2	BITS b5 b1	
	200 128 80	DCS	220 144 90	(undef)	240 160 A0	°	260 176 B0	À	300 192 C0		320 208 D0	à	340 224 E0		360 240 F0	0	0	0	0	0
	201 129 81	PU1	221 145 91	¡	241 161 A1	±	261 177 B1	Á	301 193 C1	Ñ	321 209 D1	á	341 225 E1	ñ	361 241 F1	0	0	0	1	1
	202 130 82	PU2	222 146 92	¢	242 162 A2	²	262 178 B2	Â	302 194 C2	Ò	322 210 D2	â	342 226 E2	ò	362 242 F2	0	0	1	0	2
	203 131 83	STS	223 147 93	£	243 163 A3	³	263 179 B3	Ã	303 195 C3	Ó	323 211 D3	ã	343 227 E3	ó	363 243 F3	0	0	1	1	3
IND	204 132 84	CCH	224 148 94		244 164 A4		264 180 B4	Ä	304 196 C4	Ô	324 212 D4	ä	344 228 E4	ô	364 244 F4	0	1	0	0	4
NEL	205 133 85	MW	225 149 95	¥	245 165 A5	µ	265 181 B5	Å	305 197 C5	Õ	325 213 D5	å	345 229 E5	õ	365 245 F5	0	1	0	1	5
SSA	206 134 86	SPA	226 150 96		246 166 A6	¶	266 182 B6	Æ	306 198 C6	Ö	326 214 D6	æ	346 230 E6	ö	366 246 F6	0	1	1	0	6
ESA	207 135 87	EPA	227 151 97	§	247 167 A7	·	267 183 B7	Ç	307 199 C7	Œ	327 215 D7	ç	347 231 E7	œ	367 247 F7	0	1	1	1	7
HTS	210 136 88		230 152 98	¤	250 168 A8		270 184 B8	È	310 200 C8	Ø	330 216 D8	è	350 232 E8	ø	370 248 F8	1	0	0	0	8
HTJ	211 137 89		231 153 99	©	251 169 A9	¹	271 185 B9	É	311 201 C9	Ù	331 217 D9	é	351 233 E9	ù	371 249 F9	1	0	0	1	9
VTS	212 138 8A		232 154 9A	ª	252 170 AA	º	272 186 BA	Ê	312 202 CA	Ú	332 218 DA	ê	352 234 EA	ú	372 250 FA	1	0	1	0	10
PLD	213 139 8B	CSI	233 155 9B	«	253 171 AB	»	273 187 BB	Ë	313 203 CB	Û	333 219 DB	ë	353 235 EB	û	373 251 FB	1	0	1	1	11
PLU	214 140 8C	ST	234 156 9C		254 172 AC	¼	274 188 BC	Ì	314 204 CC	Ü	334 220 DC	ì	354 236 EC	ü	374 252 FC	1	1	0	0	12
RI	215 141 8D	OSC	235 157 9D		255 173 AD	½	275 189 BD	Í	315 205 CD	Ÿ	335 221 DD	í	355 237 ED	ÿ	375 253 FD	1	1	0	1	13
SS2	216 142 8E	PM	236 158 9E		256 174 AE		276 190 BE	Î	316 206 CE		336 222 DE	î	356 238 EE		376 254 FE	1	1	1	0	14
SS3	217 143 8F	APC	237 159 9F		257 175 AF	¿	277 191 BF	Ï	317 207 CF	ß	337 223 DF	ï	357 239 EF	(undef)	377 255 FF	1	1	1	1	15
		C1 CODES				GR CODES (DEC SUPPLEMENTAL GRAPHICS)														

Table 2-4: DEC Special Graphics Character Set

ROW	COLUMN b4	b3	b2	b1	0		1		2		3		4		5		6		7	
	BITS b8 / b7 / b6 / b5				0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	◆	140 96 60	⎺ SCAN 3	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	▒	141 97 61	⎻ SCAN 5	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	HT	142 98 62	⎼ SCAN 7	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	FF	143 99 63	⎽ SCAN 9	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	CR	144 100 64	├	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	LF	145 101 65	┤	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	°	146 102 66	┴	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	±	147 103 67	┬	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	NL	150 104 68	│	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	VT	151 105 69	≤	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	┘	152 106 6A	≥	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	[	133 91 5B	┐	153 107 6B	π	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	\	134 92 5C	┌	154 108 6C	≠	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	]	135 93 5D	└	155 109 6D	£	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	┼	156 110 6E	·	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	(blank)	137 95 5F	⎯ SCAN 1	157 111 6F	DEL	177 127 7F
					C0 CODES				GL CODES (DEC SPECIAL GRAPHICS)											

KEY		
CHARACTER	ESC	33 27 1B
		OCTAL DECIMAL HEX

Table 2-5: British NRC Set (British Keyboard Selection)

ROW	COLUMN b4	b3	b2	b1	0		1		2		3		4		5		6		7	
BITS b8					0		0		0		0		0		0		0		0	
b7					0		0		0		0		1		1		1		1	
b6					0		0		1		1		0		0		1		1	
b5					0		1		0		1		0		1		0		1	
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	£	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	[	133 91 5B	k	153 107 6B	{	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	\	134 92 5C	l	154 108 6C	|	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	]	135 93 5D	m	155 109 6D	}	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	~	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY

CHARACTER			
	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-6: Dutch NRC Set (Dutch Keyboard Selection)

ROW	COLUMN				0		1		2		3		4		5		6		7	
	b8	b7	b6	b5 BITS	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	¾	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	£	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	ij	133 91 5B	k	153 107 6B	¨	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	½	134 92 5C	l	154 108 6C	f	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	|	135 93 5D	m	155 109 6D	¼	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	´	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-7: Finnish NRC Set (Finnish Keyboard Selection)

ROW	COLUMN				0		1		2		3		4		5		6		7	
	b8			BITS	0		0		0		0		0		0		0		0	
		b7			0		0		0		0		1		1		1		1	
			b6		0		0		1		1		0		0		1		1	
				b5	0		1		0		1		0		1		0		1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	é	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	Ä	133 91 5B	k	153 107 6B	ä	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	Ö	134 92 5C	l	154 108 6C	ö	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	Å	135 93 5D	m	155 109 6D	å	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	Ü	136 94 5E	n	156 110 6E	ü	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY

CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-8: French NRC Set (Flemish and French/Belgian Keyboard Selections)

ROW	COLUMN				0		1		2		3		4		5		6		7	
	b8	b7	b6	b5 BITS	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	à	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	£	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	°	133 91 5B	k	153 107 6B	é	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	ç	134 92 5C	l	154 108 6C	ù	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	§	135 93 5D	m	155 109 6D	è	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	¨	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-9: French Canadian NRC Set (French Canadian Keyboard Selection)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	à	100 64 40	P	120 80 50	ô	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	â	133 91 5B	k	153 107 6B	é	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	ç	134 92 5C	l	154 108 6C	ù	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	ê	135 93 5D	m	155 109 6D	è	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	î	136 94 5E	n	156 110 6E	û	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-10: German NRC Set (German Keyboard Selection)

ROW	COLUMN				0		1		2		3		4		5		6		7	
	b8 b7 b6 b5 BITS				0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	§	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	Ä	133 91 5B	k	153 107 6B	ä	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	Ö	134 92 5C	l	154 108 6C	ö	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	Ü	135 93 5D	m	155 109 6D	ü	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	ß	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-11: Italian NRC Set (Italian Keyboard Selection)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	§	100 64 40	P	120 80 50	ù	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	£	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	°	133 91 5B	k	153 107 6B	à	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	ç	134 92 5C	l	154 108 6C	ò	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	é	135 93 5D	m	155 109 6D	è	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	ì	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-12: Norwegian/Danish NRC Set (Danish and Norwegian Keyboard Selections)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	Ä	100 64 40	P	120 80 50	ä	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	Æ	133 91 5B	k	153 107 6B	æ	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	Ø	134 92 5C	l	154 108 6C	ø	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	Å	135 93 5D	m	155 109 6D	å	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	Ü	136 94 5E	n	156 110 6E	ü	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-13: Spanish NRC Set (Spanish Keyboard Selection)

ROW	COLUMN				0		1		2		3		4		5		6		7	
	b8	b7	b6	b5 BITS	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	§	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	£	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	¡	133 91 5B	k	153 107 6B	°	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	Ñ	134 92 5C	l	154 108 6C	ñ	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	¿	135 93 5D	m	155 109 6D	ç	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	~	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY

CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-14: Swedish NRC Set (Swedish Keyboard Selection)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	É	100 64 40	P	120 80 50	é	140 96 60	p	160 112 70
1	0	0	0	1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	ETX	3 3 3	DC3 (XOFF)	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	Ä	133 91 5B	k	153 107 6B	ä	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	Ö	134 92 5C	l	154 108 6C	ö	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	Å	135 93 5D	m	155 109 6D	å	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	Ü	136 94 5E	n	156 110 6E	ü	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY

CHARACTER			
	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-15: Swiss NRC Set (Swiss/French and Swiss/German Keyboard Selections)

ROW	COLUMN / BITS b8 b7 b6 b5 → b4 b3 b2 b1	0		1		2		3		4		5		6		7	
		0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
0	0 0 0 0	NUL	0 0 0	DLE	20 16 10	SP	40 32 20	0	60 48 30	à	100 64 40	P	120 80 50	ô	140 96 60	p	160 112 70
1	0 0 0 1	SOH	1 1 1	DC1 (XON)	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0 0 1 0	STX	2 2 2	DC2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0 0 1 1	ETX	3 3 3	DC3 (XOFF)	23 19 13	ù	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0 1 0 0	EOT	4 4 4	DC4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0 1 0 1	ENQ	5 5 5	NAK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0 1 1 0	ACK	6 6 6	SYN	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0 1 1 1	BEL	7 7 7	ETB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1 0 0 0	BS	10 8 8	CAN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1 0 0 1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1 0 1 0	LF	12 10 A	SUB	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1 0 1 1	VT	13 11 B	ESC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	é	133 91 5B	k	153 107 6B	ä	173 123 7B
12	1 1 0 0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	ç	134 92 5C	l	154 108 6C	ö	174 124 7C
13	1 1 0 1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	ê	135 93 5D	m	155 109 6D	ü	175 125 7D
14	1 1 1 0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	î	136 94 5E	n	156 110 6E	û	176 126 7E
15	1 1 1 1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	è	137 95 5F	o	157 111 6F	DEL	177 127 7F

KEY			
CHARACTER	ESC	33 27 1B	OCTAL DECIMAL HEX

Table 2-16: Display Controls Font (Sheet 1 | 2)

ROW	COLUMN			BITS	0		1		2		3		4		5		6		7	
	b8	b7	b6	b5	0 0 0 0		0 0 0 1		0 0 1 0		0 0 1 1		0 1 0 0		0 1 0 1		0 1 1 0		0 1 1 1	
	b4	b3	b2	b1																
0	0	0	0	0	NU	0 0 0	DL	20 16 10		40 32 20	0	60 48 30	@	100 64 40	P	120 80 50	`	140 96 60	p	160 112 70
1	0	0	0	1	SH	1 1 1	D1	21 17 11	!	41 33 21	1	61 49 31	A	101 65 41	Q	121 81 51	a	141 97 61	q	161 113 71
2	0	0	1	0	SX	2 2 2	D2	22 18 12	"	42 34 22	2	62 50 32	B	102 66 42	R	122 82 52	b	142 98 62	r	162 114 72
3	0	0	1	1	EX	3 3 3	D3	23 19 13	#	43 35 23	3	63 51 33	C	103 67 43	S	123 83 53	c	143 99 63	s	163 115 73
4	0	1	0	0	ET	4 4 4	D4	24 20 14	$	44 36 24	4	64 52 34	D	104 68 44	T	124 84 54	d	144 100 64	t	164 116 74
5	0	1	0	1	EQ	5 5 5	NK	25 21 15	%	45 37 25	5	65 53 35	E	105 69 45	U	125 85 55	e	145 101 65	u	165 117 75
6	0	1	1	0	AK	6 6 6	SY	26 22 16	&	46 38 26	6	66 54 36	F	106 70 46	V	126 86 56	f	146 102 66	v	166 118 76
7	0	1	1	1	BL	7 7 7	EB	27 23 17	'	47 39 27	7	67 55 37	G	107 71 47	W	127 87 57	g	147 103 67	w	167 119 77
8	1	0	0	0	BS	10 8 8	CN	30 24 18	(	50 40 28	8	70 56 38	H	110 72 48	X	130 88 58	h	150 104 68	x	170 120 78
9	1	0	0	1	HT	11 9 9	EM	31 25 19	)	51 41 29	9	71 57 39	I	111 73 49	Y	131 89 59	i	151 105 69	y	171 121 79
10	1	0	1	0	LF	12 10 A	⸮	32 26 1A	*	52 42 2A	:	72 58 3A	J	112 74 4A	Z	132 90 5A	j	152 106 6A	z	172 122 7A
11	1	0	1	1	VT	13 11 B	EC	33 27 1B	+	53 43 2B	;	73 59 3B	K	113 75 4B	[	133 91 5B	k	153 107 6B	{	173 123 7B
12	1	1	0	0	FF	14 12 C	FS	34 28 1C	,	54 44 2C	<	74 60 3C	L	114 76 4C	\	134 92 5C	l	154 108 6C	|	174 124 7C
13	1	1	0	1	CR	15 13 D	GS	35 29 1D	-	55 45 2D	=	75 61 3D	M	115 77 4D	]	135 93 5D	m	155 109 6D	}	175 125 7D
14	1	1	1	0	SO	16 14 E	RS	36 30 1E	.	56 46 2E	>	76 62 3E	N	116 78 4E	^	136 94 5E	n	156 110 6E	~	176 126 7E
15	1	1	1	1	SI	17 15 F	US	37 31 1F	/	57 47 2F	?	77 63 3F	O	117 79 4F	_	137 95 5F	o	157 111 6F	DT	177 127 7F
					C0 CODES				GL CODES (ASCII GRAPHICS)											

KEY		
CHARACTER	ESC	33 OCTAL 27 DECIMAL 1B HEX

Table 2-16: Display Controls Font (Sheet 1 l 2)

8		9		10		11		12		13		14		15		COLUMN				ROW
1 0 0 0		1 0 0 1		1 0 1 0		1 0 1 1		1 1 0 0		1 1 0 1		1 1 1 0		1 1 1 1		b8 b4	b7 b3	b6 b2	BITS b5 b1	
8_0	200 128 80	9_0	220 144 90	A_0	240 160 A0	°	260 176 B0	À	300 192 C0	D_0	320 208 D0	à	340 224 E0	F_0	360 240 F0	0	0	0	0	0
8_1	201 129 81	9_1	221 145 91	¡	241 161 A1	±	261 177 B1	Á	301 193 C1	Ñ	321 209 D1	á	341 225 E1	ñ	361 241 F1	0	0	0	1	1
8_2	202 130 82	9_2	222 146 92	¢	242 162 A2	²	262 178 B2	Â	302 194 C2	Ò	322 210 D2	â	342 226 E2	ò	362 242 F2	0	0	1	0	2
8_3	203 131 83	9_3	223 147 93	£	243 163 A3	³	263 179 B3	Ã	303 195 C3	Ó	323 211 D3	ã	343 227 E3	ó	363 243 F3	0	0	1	1	3
8_4	204 132 84	9_4	224 148 94	A_4	244 164 A4	B_4	264 180 B4	Ä	304 196 C4	Ô	324 212 D4	ä	344 228 E4	ô	364 244 F4	0	1	0	0	4
8_5	205 133 85	9_5	225 149 95	¥	245 165 A5	µ	265 181 B5	Å	305 197 C5	Õ	325 213 D5	å	345 229 E5	õ	365 245 F5	0	1	0	1	5
8_6	206 134 86	9_6	226 150 96	A_6	246 166 A6	¶	266 182 B6	Æ	306 198 C6	Ö	326 214 D6	æ	346 230 E6	ö	366 246 F6	0	1	1	0	6
8_7	207 135 87	9_7	227 151 97	§	247 167 A7	·	267 183 B7	Ç	307 199 C7	Œ	327 215 D7	ç	347 231 E7	œ	367 247 F7	0	1	1	1	7
8_8	210 136 88	9_8	230 152 98	¤	250 168 A8	B_8	270 184 B8	È	310 200 C8	Ø	330 216 D8	è	350 232 E8	ø	370 248 F8	1	0	0	0	8
8_9	211 137 89	9_9	231 153 99	©	251 169 A9	¹	271 185 B9	É	311 201 C9	Ù	331 217 D9	é	351 233 E9	ù	371 249 F9	1	0	0	1	9
8_A	212 138 8A	9_A	232 154 9A	ª	252 170 AA	º	272 186 BA	Ê	312 202 CA	Ú	332 218 DA	ê	352 234 EA	ú	372 250 FA	1	0	1	0	10
8_B	213 139 8B	9_B	233 155 9B	«	253 171 AB	»	273 187 BB	Ë	313 203 CB	Û	333 219 DB	ë	353 235 EB	û	373 251 FB	1	0	1	1	11
8_C	214 140 8C	9_C	234 156 9C	A_C	254 172 AC	¼	274 188 BC	Ì	314 204 CC	Ü	334 220 DC	ì	354 236 EC	ü	374 252 FC	1	1	0	0	12
8_D	215 141 8D	9_D	235 157 9D	A_D	255 173 AD	½	275 189 BD	Í	315 205 CD	Ÿ	335 221 DD	í	355 237 ED	ÿ	375 253 FD	1	1	0	1	13
8_E	216 142 8E	9_E	236 158 9E	A_E	256 174 AE	B_E	276 190 BE	Î	316 206 CE	D_E	336 222 DE	î	356 238 EE	F_E	376 254 FE	1	1	1	0	14
8_F	217 143 8F	9_F	237 159 9F	A_F	257 175 AF	¿	277 191 BF	Ï	317 207 CF	ß	337 223 DF	ï	357 239 EF	□	377 255 FF	1	1	1	1	15
		C1 CODES				GR CODES (DEC SUPPLEMENTAL GRAPHICS)														

Table 3-1 Codes Generated by Editing Keys

	Code Generated	
Key	**VT200 Mode**	**VT100, VT52 Modes**
FIND	9/11 3/1 7/14 CSI 1 ~	--
INSERT HERE	9/11 3/2 7/14 CSI 2 ~	--
REMOVE	9/11 3/3 7/14 CSI 3 ~	--
SELECT	9/11 3/4 7/14 CSI 4 ~	--
PREV SCREEN	9/11 3/5 7/14 CSI 5 ~	--
NEXT SCREEN	9/11 3/6 7/14 CSI 6 ~	--
In VT100 or VT52 modes the editing keys do not generate codes.		

Table 3-2 Codes Generated by Cursor Control Keys

	ANSI Mode*		VT52 Mode*	
Key	**Cursor Key Mode Reset Normal**	**Cursor Key Mode Set (Application)**	**Normal**	**Application**
↑	9/11 4/1 CSI A	8/15 4/1 SS3 A	1/11 4/1 ESC A	1/11 4/1 ESC A
↓	9/11 4/2 CSI B	8/15 4/2 SS3 B	1/11 4/2 ESC B	1/11 4/2 ESC B
→	9/11 4/3 CSI C	8/15 4/3 SS3 C	1/11 4/3 ESC C	1/11 4/3 ESC C
←	9/11 4/4 CSI D	8/15 4/4 SS3 D	1/11 4/4 ESC D	1/11 4/4 ESC D

* ANSI mode applies to VT200 and VT100 modes. VT52 mode is an ANSI-incompatible mode.

Table 3-3 Codes Generated by Auxiliary Keypad Keys

	VT100/VT200 ANSI Mode*		VT52 Mode*	
Key	**Keypad Numeric Mode**	**Keypad Application Mode**	**Keypad Numeric Mode**	**Keypad Application Mode**
0	3/0 0	8/15 7/0 SS3 p	3/0 0	1/11 3/15 7/0 ESC ? p
1	3/1 1	8/15 7/1 SS3 q	3/1 1	1/11 3/15 7/1 ESC ? q
2	3/2 2	8/15 7/2 SS3 r	3/2 2	1/11 3/15 7/2 ESC ? r
3	3/3 3	8/15 7/3 SS3 s	3/3 3	1/11 3/15 7/3 ESC ? s
4	3/4 4	8/15 7/4 SS3 t	3/4 4	1/11 3/15 7/4 ESC ? t
5	3/5 5	8/15 7/5 SS3 u	3/5 5	1/11 3/15 7/5 ESC ? u
6	3/6 6	8/15 7/6 SS3 v	3/6 6	1/11 3/15 7/6 ESC ? v
7	3/7 7	8/15 7/7 SS3 w	3/7 7	1/11 3/15 7/7 ESC ? w
8	3/8 8	8/15 7/8 SS3 x	3/8 8	1/11 3/15 7/8 ESC ? x
9	3/9 9	8/15 7/9 SS3 y	3/9 9	1/11 3/15 7/9 ESC ? y
-	2/13 – (minus)	8/15 6/13 SS3 m	2/13 –	1/11 3/15 6/13† ESC ? m
,	2/12 , (comma)	8/15 6/12 SS3 l	2/12 ,	1/11 3/15 6/12† ESC ? l
.	2/14 . (period)	8/15 6/14 SS3 n	2/14 .	1/11 3/15 6/14 ESC ? n
Enter‡	0/13 CR	8/15 4/13 SS3 M	0/13 CR	1/11 3/15 4/13 ESC ? M
or	0/13 0/10 CR LF		or 0/13 0/10 CR LF	
PF1	8/15 5/0 SS3 P	8/15 5/0 SS3 P	1/11 5/0 ESC P	1/11 5/0 ESC P
PF2	8/15 5/1 SS3 Q	8/15 5/1 SS3 Q	1/11 5/1 ESC Q	1/11 5/1 ESC Q
PF3	8/15 5/2 SS3 R	8/15 5/2 SS3 R	1/11 5/2 ESC R	1/11 5/2 ESC R
PF4	8/15 5/3 SS3 S	8/15 5/3 SS3 S	1/11 5/3 ESC S	1/11 5/3‡ ESC S

* ANSI mode applies to VT200 and VT100 modes. VT52 mode is an ANSI-incompatible mode.

† You cannot generate these sequences on a VT52 terminal.

‡ Keypad Numeric Mode. ENTER generates the same codes as RETURN. You can change the code generated by RETURN with the Linefeed/New Line Mode. When reset, the Linefeed/New Line Mode causes RETURN to generate a single control character (CR). When set, the mode causes RETURN to generate two control characters (CR, LF).

Table 3-4 Codes Generated by Top Row Function Keys

		Code Generated				
Name on Legend Strip	**Generic Name**	**VT200 Mode**				**VT100, VT52 Modes**
HOLD SCREEN	(F1)*	--				--
PRINT SCREEN	(F2)*	--				--
SET-UP	(F3)*	--				--
DATA/TALK	(F4)*	--				--
BREAK	(F5)*	--				--
F6	F6	9/11 CSI	3/1 1	3/7 7	7/14 ~	--
F7	F7	9/11 CSI	3/1 1	3/8 8	7/14 ~	--
F8	F8	9/11 CSI	3/1 1	3/9 9	7/14 ~	--
F9	F9	9/11 CSI	3/2 2	3/0 0	7/14 ~	--
F10	F10	9/11 CSI	3/2 2	3/1 1	7/14 ~	--
F11 (ESC)	F11	9/11 CSI	3/2 2	3/3 3	7/14 ~	1/11 ESC
F12 (BS)	F12	9/11 CSI	3/2 2	3/4 4	7/14 ~	0/8 BS
F13 (LF)	F13	9/11 CSI	3/2 2	3/5 5	7/14 ~	0/10 LF
F14	F14	9/11 CSI	3/2 2	3/6 6	7/14 ~	--
HELP	(F15)	9/11 CSI	3/2 2	3/8 8	7/14 ~	--
DO	(F16)	9/11 CSI	3/2 2	3/9 9	7/14 ~	--
F17	F17	9/11 CSI	3/3 3	3/1 1	7/14 ~	--
F18	F18	9/11 CSI	3/3 3	3/2 2	7/14 ~	--
F19	F19	9/11 CSI	3/3 3	3/3 3	7/14 ~	--
F20	F20	9/11 CSI	3/3 3	3/4 4	7/14 ~	--

* F1 through F5 are local function keys and do not generate codes.

Table 3-5 Keys Used to Generate 7-Bit Control Characters

Control Character Mnemonic	Code	Key Pressed with CTRL (All Modes)	Dedicated Function Key
NUL	0/00	2, space	
SOH	0/01	A	
STX	0/02	B	
ETX	0/03	C	
EOT	0/04	D	
ENQ	0/05	E	
ACK	0/06	F	
BEL	0/07	G	
BS	0/08	H	F12 (BS)*
HT	0/09	I	TAB
LF	0/10	J	F13 (LF)*
VT	0/11	K	
FF	0/12	L	
CR	0/13	M	RETURN
SO	0/14	N	
SI	0/15	O	
DLE	1/00	P	
DC1	1/01	Q**	
DC2	1/02	R	
DC3	1/03	S**	
DC4	1/04	T	
NAK	1/05	U	
SYN	1/06	V	
ETB	1/07	W	
CAN	1/08	X	
EM	1/09	Y	
SUB	1/10	Z	
ESC	1/11	3,[	F11 (ESC)*
FS	1/12	4,/	
GS	1/13	5,]	
RS	1/14	6,~	
US	1/15	7,?	
DEL	7/15	8	DELETE

* Keys F11, F12, and F13 generate these 7-bit control characters only when the terminal is operated in VT100 mode or VT52 mode.

** These keystrokes are enabled only if XOFF support is disabled. If XOFF support is enabled, then CTRL-S is a "hold screen" local function and CTRL-Q is an "unhold screen" local function.

Table 4-1 C0 (ASCII) Control Characters Recognized

Mnemonic	Code	Name	Action Taken
NUL	0/0	Null	Ignored when received.
ENQ	0/5	Enquiry	Answerback message is generated.
BEL	0/7	Bell	Generates bell tone if bell is enabled.
BS	0/8	Backspace	Moves cursor to the left one character position: if cursor is at left margin, no action occurs.
HT	0/9	Horizontal tabulation	Moves cursor to next tab stop, or to right margin if there are no more tab stops. Does not cause autowrap.
LF	0/10	Linefeed	Causes a linefeed or a new line operation, depending on the setting of new line mode.
VT	0/11	Vertical tabulation	Processed as LF.
FF	0/12	Form feed	Processed as LF.
CR	0/13	Carriage return	Moves cursor to left margin on current line.
SO (LS1)	0/14	Shift out (Lock shift G1)	Invokes G1 character set into GL. G1 is designated by a select-character-set (SCS) sequence.
SI (LS0)	0/15	Shift in (Lock shift G0)	Invoke G0 character set into GL. G0 is designated by a select-character-set sequence (SCS).
DC1	1/1	Device Control 1	Also referred to as XON. If XOFF support is enabled, DC1 clears DC3 (XOFF), causing the terminal to continue transmitting characters (keyboard unlocks) unless KAM mode is currently set.
DC3	1/3	Device Control 3	Also referred to as XOFF. If XOFF support is enabled, DC3 causes the terminal to stop transmitting characters until a DC1 control character is received.
CAN	1/8	Cancel	If received during an escape or control sequence, terminates and cancels the sequence. No error character is displayed. If received during a device control string, the DCS is terminated and no error character is displayed.
SUB	1/10	Substitute	If received during escape or control sequence, terminates and cancels the sequence. Causes a reverse question mark to be displayed. If received during a device control sequence, the DCS is terminated and reverse question mark is displayed.
ESC	1/11	Escape	Processed as escape sequence introducer. Terminates any escape, control or device control sequence which is in progress.
DEL	7/15	Delete	Ignored when received. Note: May not be used as a time fill character.

Table 4-2 C1 Control Characters Recognized

Mnemonic	8-Bit Code	Equivalent 7-Bit Code Extension	Name	Action Taken
IND	8/4	1/11 4/4 ESC D	Index	Moves cursor down one line in same column. If cursor is at bottom margin, screen performs a scroll up.
NEL	8/5	1/11 4/5 ESC E	Next line	Moves cursor to first position on next line. If cursor is at bottom margin, screen performs a scroll up.
HTS	8/8	1/11 4/8 ESC H	Horizontal tab set	Sets one horizontal tab stop at the column where the cursor is.
RI	8/13	1/11 4/13 ESC M	Reverse index	Moves cursor up one line in same column. If cursor is at top margin, screen performs a scroll down.
SS2	8/14	1/11 4/14 ESC N	Single shift G2	Temporarily invokes G2 character set into GL for the next graphic character. G2 is designated by a select-character-set (SCS) sequence.
SS3	8/15	1/11 4/15 ESC O	Single shift G3	Temporarily invokes G3 character set into GL for the next graphic character. G3 is designated by a select-character-set (SCS) sequence.
DCS	9/0	1/11 5/0 ESC P	Device control string	Processed as opening delimiter of a device control string for device control use.
CSI	9/11	1/11 5/11 ESC [	Control sequence introducer	Processed as control sequence introducer.
ST	9/12	1/11 5/12 ESC \	String terminator	Processed as closing delimiter of a string opened by DCS.

Table 4-3 Level 1-Level 2 Compatibility Comparison

Difference Area	**Level 1 (VT100 Mode)**	**Level 2 (VT200 Mode)**
Keyboard	Sends ASCII only. Keystrokes that normally send DEC Supplemental Characters transfer nothing. (Interpreted as an error.) User-defined keys are do not operate. Special function keys and six editing keys do not operate (except F11, F12, and F13, which send ESC, BS, and LF, respectively).	Permits full use of VT220 keyboard.
7 or 8 bits	The 8th bit of all received characters is set to zero (0).	The 8th bit of all received characters is significant.
Character Sets	Only ASCII, national replacement characters, and DEC special graphics are available.	All VT220 character sets are available.
C1 Control Characters	All transmitted C1 controls are forced to S7C1 state and sent as 7-bit escape sequences.	–

Table 4-4 Designating Hard Character Sets

Character Set	Final Character
ASCII	4/2 B
DEC supplemental (VT200 mode only)	3/12 <
DEC special graphics	3/0 0

National replacement character sets

NOTE: Only one national character set is available at any one time (national mode).

Character Set	Final Character
British	4/1 A
Dutch	3/4 4
Finnish	4/3 C or 3/5 5
French	5/2 R
French Canadian	5/1 Q
German	4/11 K
Italian	5/9 Y
Norwegian/Danish	4/5 E or 3/6 6
Spanish	5/10 Z
Swedish	4/8 H or 3/7 7
Swiss	3/13 =

Table 4-5 Invoking Character Sets Using Lock Shifts

Control Name	Coding		Function
LS0 (lock shift G0)	0/15 SI		Invoke G0 into GL. (default)
LS1 (lock shift G1)	0/14 SO		Invoke G1 into GL.
LS1R (lock shift G1, right)	1/11 ESC	7/14 ~	Invoke G1 into GR (VT200 mode only). This sequence can cause software incompatibility problems.
LS2 (lock shift G2)	1/11 ESC	6/14 n	Invoke G2 into GL (VT200 mode only). This sequence can cause software incompatibility problems.
LS2R (lock shift G2, right)	1/11 ESC	7/13 }	Invoke G2 into GR (VT200 mode only). (default)
LS3 (lock shift G3)	1/11 ESC	6/15 o	Invoke G3 into GL (VT200 mode only). This sequence can cause software incompatibility problems.
LS3R (lock shift G3, right)	1/11 ESC	7/12 |	Invoke G3 into GR (VT200 mode only).

Table 4-6 Selectable Modes Summary

Name	Mnemonic	Set Mode	Reset Mode*
Keyboard action†	KAM	Locked `CSI 2 h`	Unlocked `CSI 2 l`
Insert/ replace	IRM	Insert `CSI 4 h`	Replace `CSI 4 l`
Send/ receive	SRM	Off `CSI 12 h`	On `CSI 12 l`
Line feed/ new line	LNM	New Line `CSI 20 h`	Line Feed `CSI 20 l`
Cursor key	DECCKM	Application `CSI ? 1 h`	Cursor `CSI ? 1 l`
ANSI/VT52	DECANM	N/A	VT52 `CSI ? 2 l`
Column	DECCOLM	132 Column `CSI ? 3 h`	80 Column `CSI ? 3 l`
Scrolling†	DECSCLM	Smooth `CSI ? 4 h`	Jump `CSI ? 4 l`
Screen†	DECSCNM	Reverse `CSI ? 5 h`	Normal `CSI ? 5 l`
Origin	DECOM	Origin `CSI ? 6 h`	Absolute `CSI ? 6 l`
Auto wrap	DECAWN	On `CSI ? 7 h`	Off `CSI ? 7 l`
Auto repeat†	DECARM	On `CSI ? 8 h`	Off `CSI ? 8 l`
Print form feed	DECPFF	On `CSI ? 18 h`	Off `CSI ? 18 l`
Print extent	DECPEX	Full Screen `CSI ? 19 h`	Scrolling Region `CSI ? 19 l`
Text cursor enable	DECTCEM	On `CSI ? 25 h`	Off `CSI ? 25 l`
Keypad	DECKPAM DECKPNM	Application `ESC =`	Numeric `ESC >`
Character set	DECNRCM	National `CSI ? 42 h`	Multinational `CSI ? 42 l`

* The last character of each sequence is lowercase L (6/12).

† User preference feature

Table 4-7 ANSI-Standardized Modes

Name	Mnemonic	Parameter (Ps)
Error (ignored)	–	0 (3/0)
Keyboard action	KAM	2 (3/2)
Insert/replace	IRM	4 (3/4)
Send/receive	SRM	12 (3/1 3/2)
Line feed/new line	LNM	20 (3/2 3/0)

Table 4-8 ANSI-Compatible DEC Private Modes

Name	Mnemonic	Parameter (Ps)
Error (ignored)	–	0 (3/0)
Cursor key	DECCKM	1 (3/1)
ANSI/VT52	DECANM	2 (3/2)
Column	DECCOLM	3 (3/3)
Scroll	DECSCLM	4 (3/4)
Screen	DECSCNM	5 (3/5)
Origin	DECOM	6 (3/6)
Auto wrap	DECAWM	7 (3/7)
Auto repeat	DECARM	8 (3/8)
Printer form feed	DECPFF	18 (3/1 3/8)
Printer extent	DECPEX	19 (3/1 3/9)
Text cursor enable	DECTCEM	25 (3/2 3/5)
National replacement character sets	DECNRCM	42 (3/4 3/2)

Table 4-9 DECDLD Parameter Characters

Parameter	Name	Description
Pfn	Font number	Selects the DRCS font buffer to load. The VT220 has only one DRCS font buffer. This parameter has two valid values, 0 and 1.
Pcn	Starting character number	Selects starting character to load in DRCS font buffer. For example, parameter value 1 specifies a column 2/row 1 character; parameter 94 specifies a column 7/row 14 character (Table 2-1).
Pe	Erase control	Selects which characters are erased before loading. 0 = erase all characters in this DRCS set. 1 = erase only the characters that are being reloaded. 2 = erase all characters in all DRCS sets (this font buffer number and other font buffer numbers).
Pcms	Character matrix size	Defines the expected limit of the character matrix size. 0 = Device default (7 × 10) 1 = (not used) 2 = 5 × 10 3 = 6 × 10 4 = 7 × 10
Pw	Width attribute	Specifies the width attribute. 0 = Device default (80 columns) 1 = 80 columns 2 = 132 columns
Pt	Text/ full-cell	Allows software to treat the font as a text font or a full-cell font. 0 = Device default (text) 1 = Text 2 = Full-cell (not used) Full-cell fonts can individually address all pixels in a cell, while text fonts, in general, may not be able to individually address all pixels.

Table 4-10 Soft Terminal Reset (DECSTR) States

Sequence	State	Stored in NVR
Text cursor	On	Yes, NVR value ignored.
Insert/replace	Replace	No
Origin mode	Absolute	No
Auto wrap	Off	Yes, NVR value ignored.
Keyboard action	Unlocked	No
Keypad dode	Numeric	No
Cursor key mode	Normal	No
Top margin	1	No
Bottom margin	24	No
Multinational/ national	Multinational	Yes, NVR value ignored.
NOTE: The current mode (national or multinational) is not affected by the "Reset Terminal" field in set-up.		
Character Sets G0, G1, G2, G3 GL, GR	VT200 defaults when in VT200 mode. VT100 defaults (via set-up only) when in VT52 or VT100 mode.	No
Video character attributes	Normal	No
Selective erase attributes	Normal (erasable by DECSEL/DECSED)	No
Save Cursor State*		No
Cursor position	Home	
Character sets	VT100 or VT200 defaults (as appropriate)	
Selective erase attribute bit write state	Off	
SGR write state	Normal	
Origin mode	Normal (reset)	
Character shift (G0 to GL, G2 to GR, no shifts)	Power-up defaults	

* Applies only to later restore cursor commands (DECRC).

Table 4-11 VT52 Escape Sequences

Escape Sequence	Function
`ESC A`	Cursor up.
`ESC B`	Cursor down.
`ESC C`	Cursor right.
`ESC D`	Cursor left.
`ESC F`	Enter graphics mode.
`ESC G`	Exit graphics mode.
`ESC H`	Cursor to home.
`ESC I`	Reverse line feed.
`ESC J`	Erase to end of screen.
`ESC K`	Erase to end of line.
`ESC Y Line Column`	Direct cursor address.
`ESC Z`	Identify.
`ESC =`	Enter alternate keypad mode.
`ESC >`	Exit alternate keypad mode.
`ESC <`	Enter ANSI mode.
`ESC ^`	Enter auto print mode.
`ESC _`	Exit auto print mode.
`ESC W`	Enter printer controller mode.
`ESC X`	Exit printer controller mode.
`ESC ]`	Print screen.
`ESC V`	Print cursor line.

Figures

Figure 1-1 VT220 Video Display Terminal

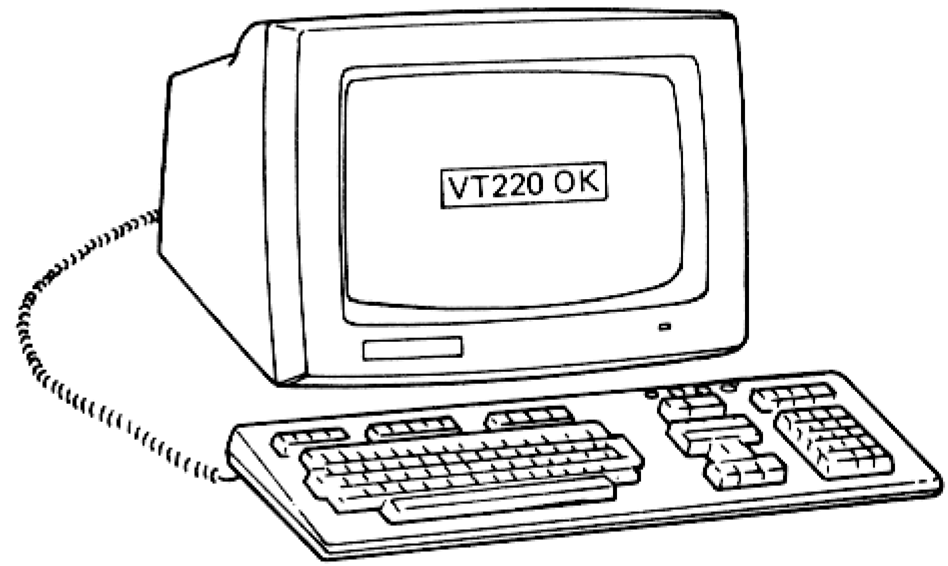


Figure 2-1 7-Bit Code

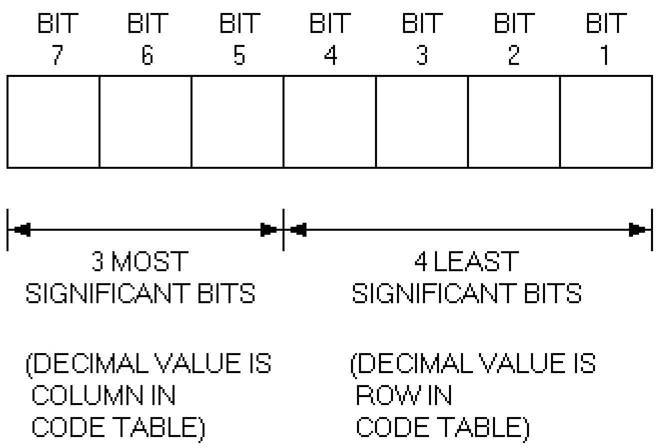


Figure 2-2 8-Bit Code

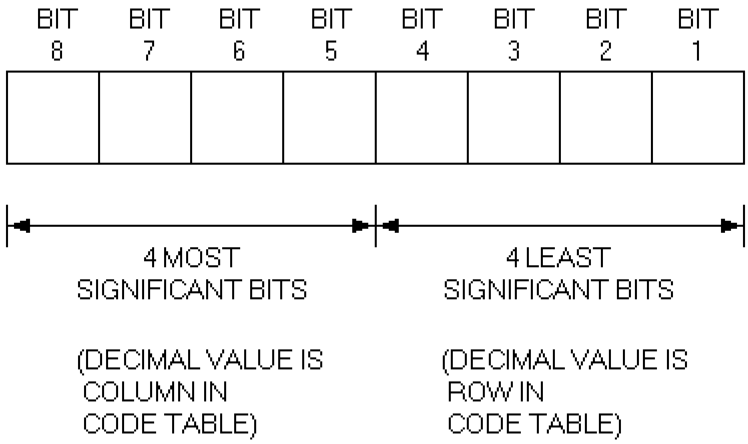


Figure 3-1 Key Groupings (North American Keyboard)

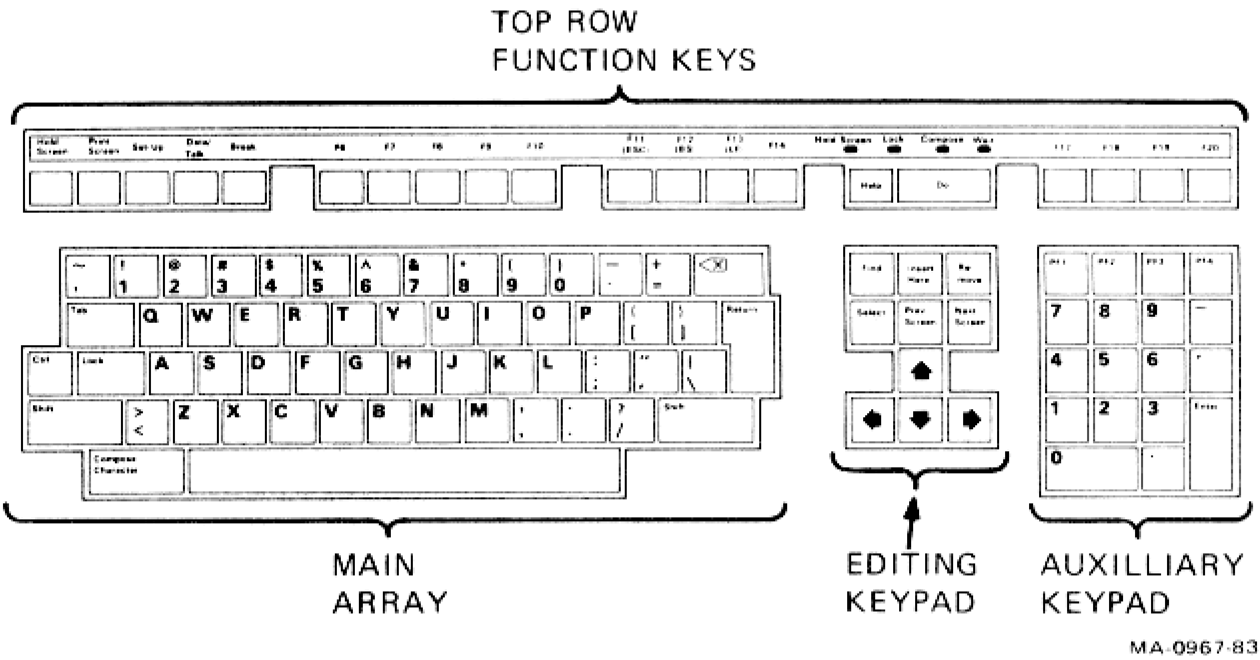
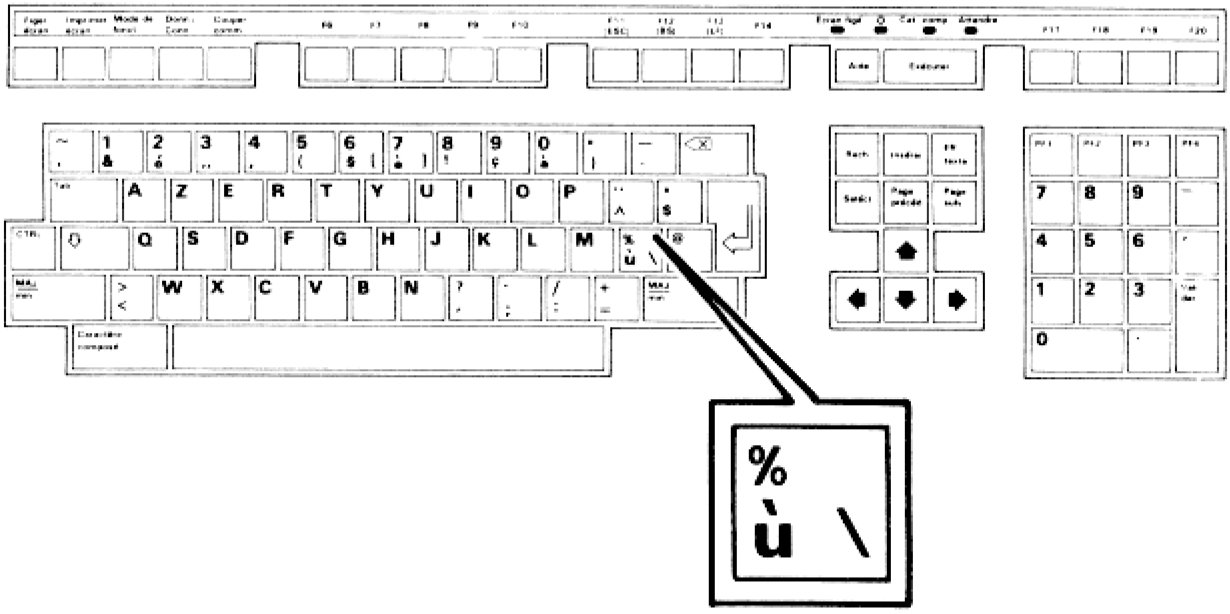


Figure 3-2 French/Belgian Keyboard



MA-0968-83

Figure 4-1 Character Set Selection

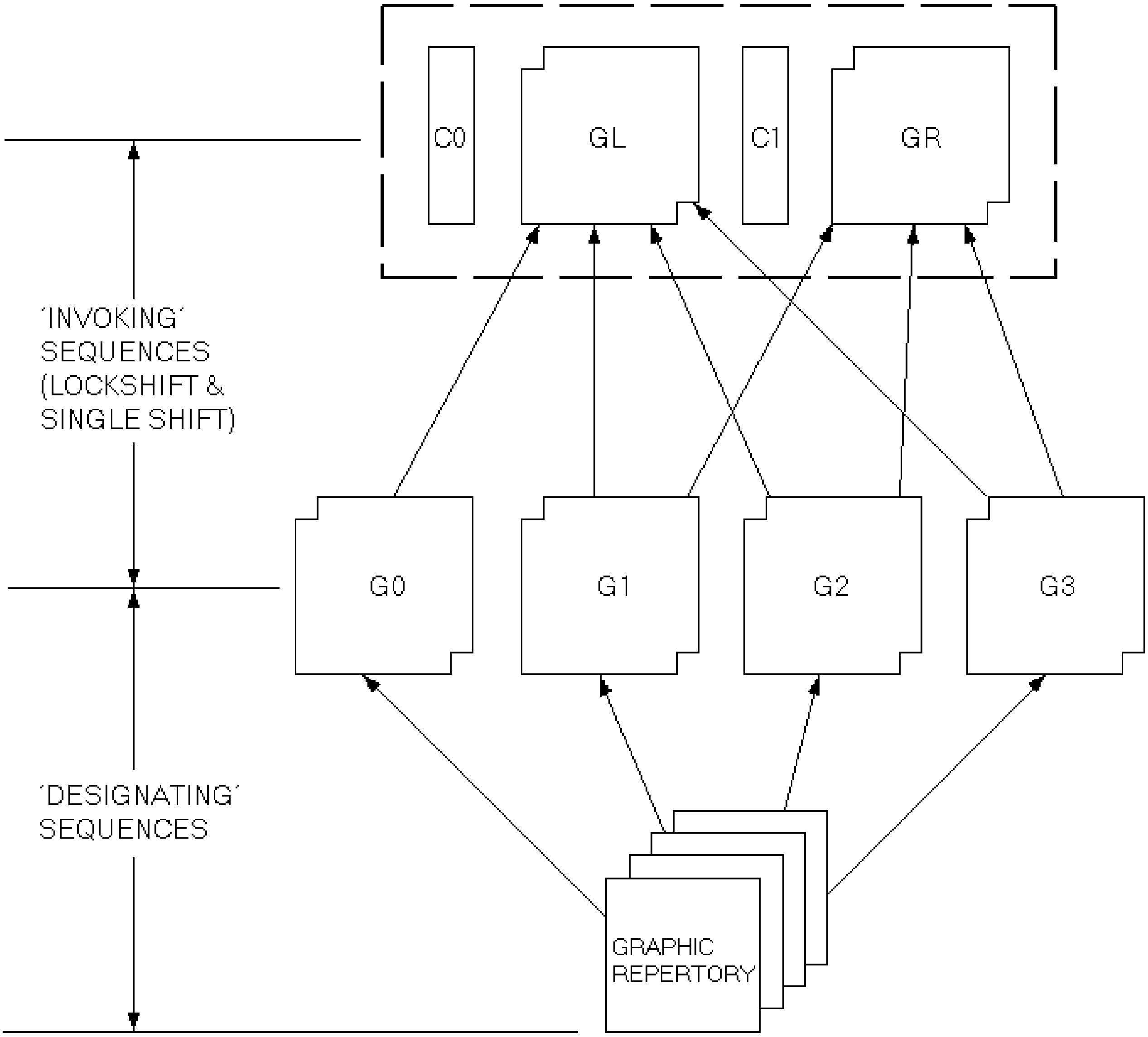


Figure 4-2 Locking and Single-Shift Commands (VT100 Mode)

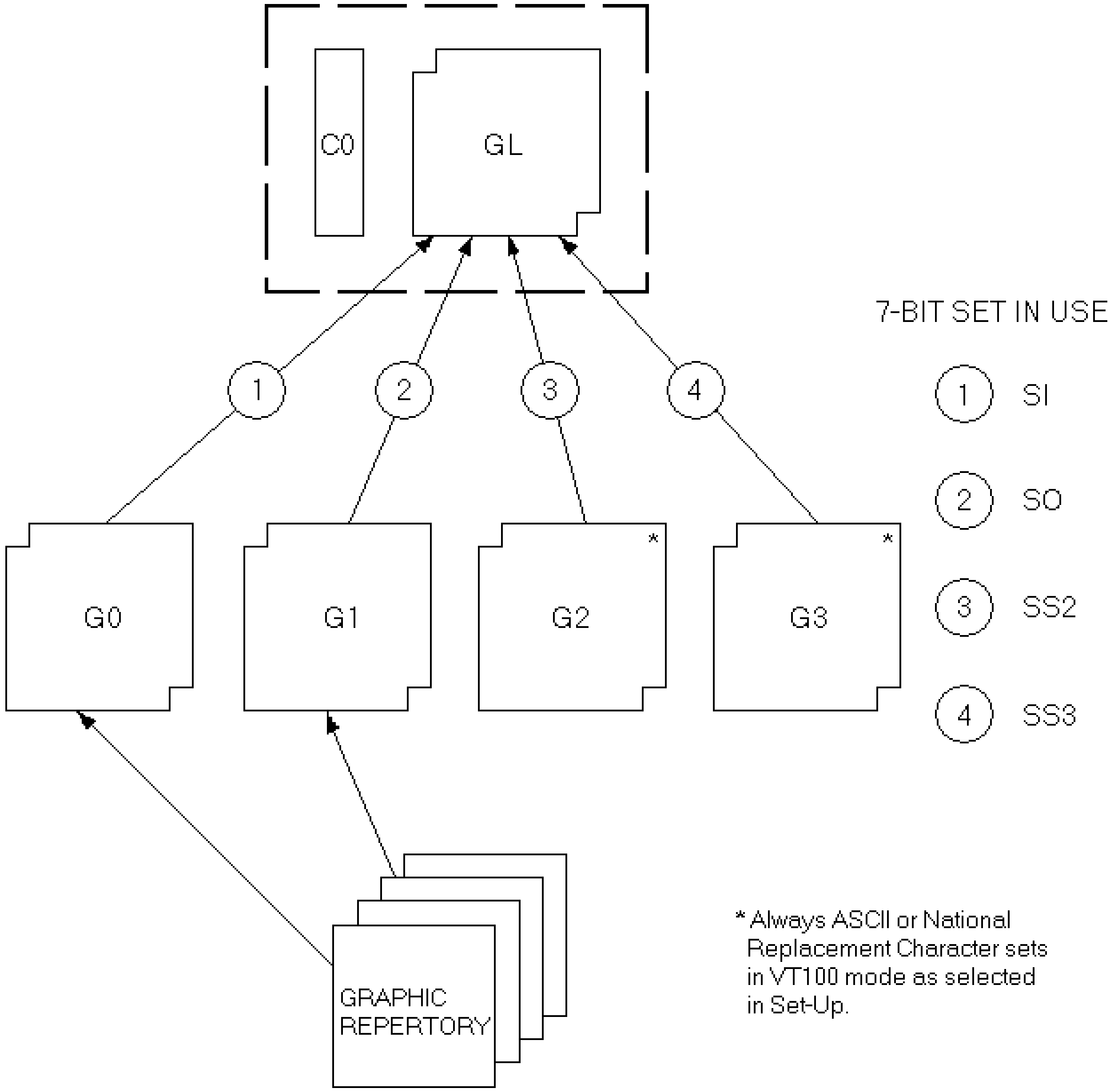


Figure 4-3 Locking and Single-Shift Commands (VT200 Mode)

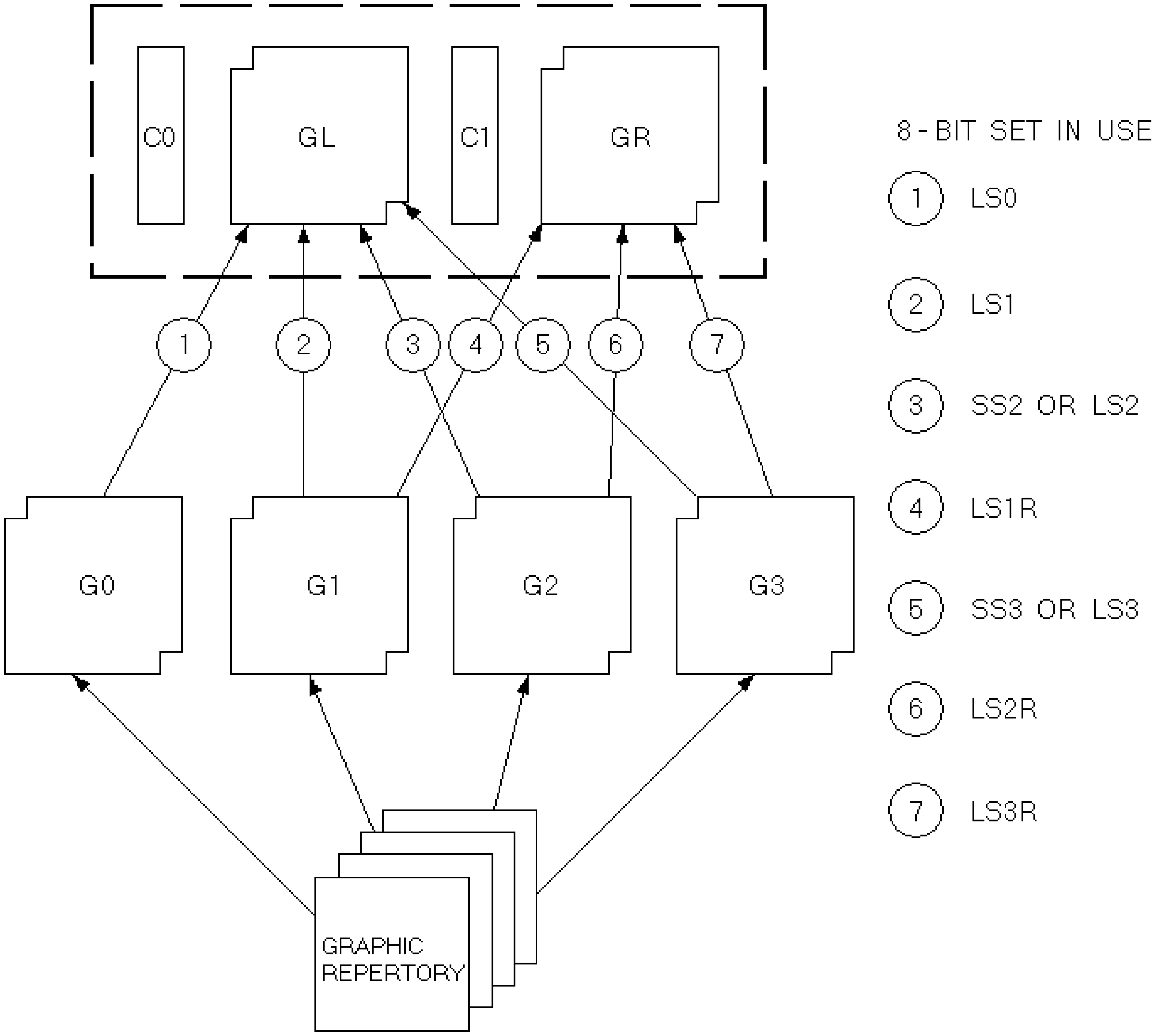


Figure 4-4 Cell and Normal Character Cell Size Relationship

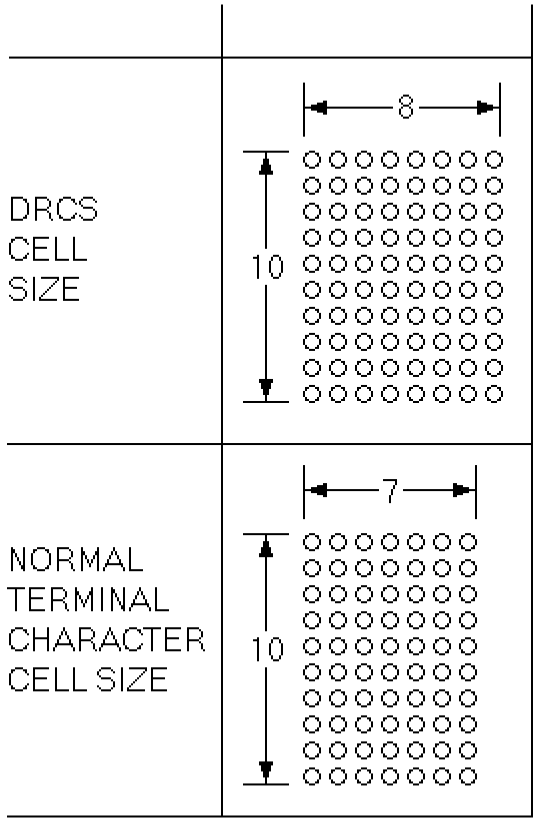


Figure 4-5 Example of an "A" Character

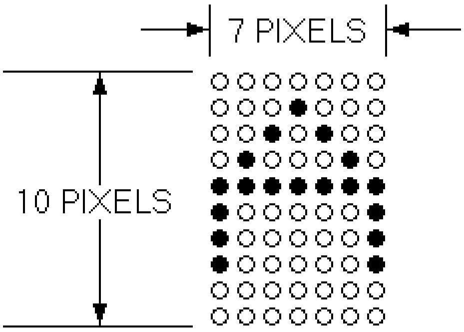


Figure 4-6 Example of "A" Divided Into Columns (6 Bits per Column)

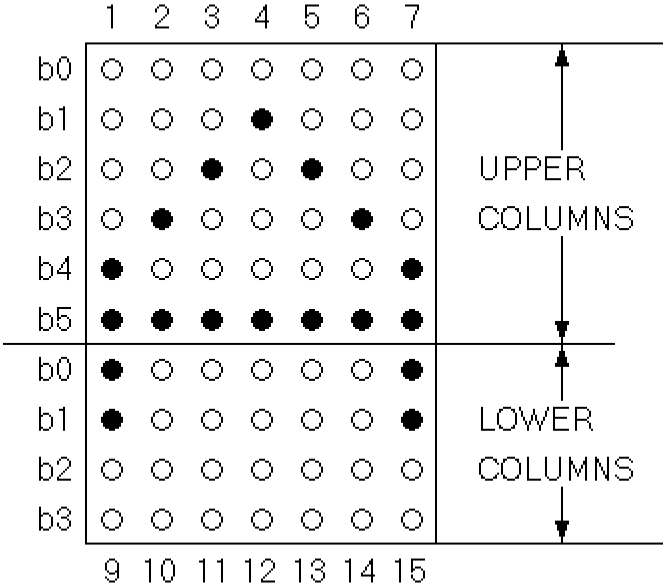


Figure 4-7 Column Codes for Example 80-column Font Character A